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Dumka Agroforestry: Year-0 Biodiversity Baseline & Monitoring Framework

Project Partner

FOF INDIA

REPORT

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17th October 2025

Executive Summary

Restoration to Resilience: The Auditable Baseline

Dumka, Jharkhand, represents a crucial blueprint for regenerative investment. Across 70 degraded sites (147.23 ha), this initiative transcends simple tree planting, establishing a living, climate-resilient economic infrastructure. Our Year-0 assessment provides a rigorous, TNFD-aligned baseline, turning the landscape's challenges into a scalable financial opportunity.

70	147.23 ha	73
Degraded Sites Under Restoration	Total Project Area	Unique Bird Species Documented

Ecological Signal: A Landscape in Recovery

Our Year-0 baseline confirms a landscape with high potential. While the starting point reflects historical degradation, key indicators point to a resilient ecosystem ready for measurable uplift and investment.

- Species Pool Strength: We documented 73 unique bird species, including the Near Threatened Alexandrine Parakeet, proving the presence of resilient faunal networks and immediate high-value habitat potential.
- Habitat Health: Initial satellite analysis shows moderate vegetation health (NDVI 0.24–0.52), but low connectivity (CPLAND 21.35%) signals that corridor restoration is the critical next step to accelerate ecological resilience by Year-3.
- Soundscape Vitality: Acoustic health indices confirm the moderate presence of robust dawn biophony (Acoustic Health = 0.496), affirming the ecosystem's foundational vitality despite human pressures.

12-24 Months Target

⇧ NDVI HHI

Enhance habitat connectivity

Raise acoustic health ≥ 3 sites

Confirm pollinator recovery

⇩ mid-day noise at hotspots

Strategic Adaptation: Turning Climate Risk into Revenue

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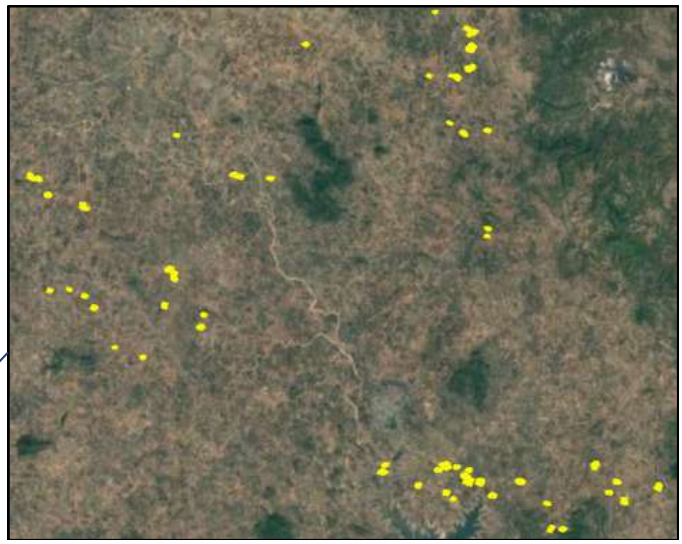
PROJECT OVERVIEW

India

Dumka, Sites



Restoration Goals:

- Re-establish native species through targeted replanting by FCF.

Arjun

Mango

Sonajhuri

Sagwan

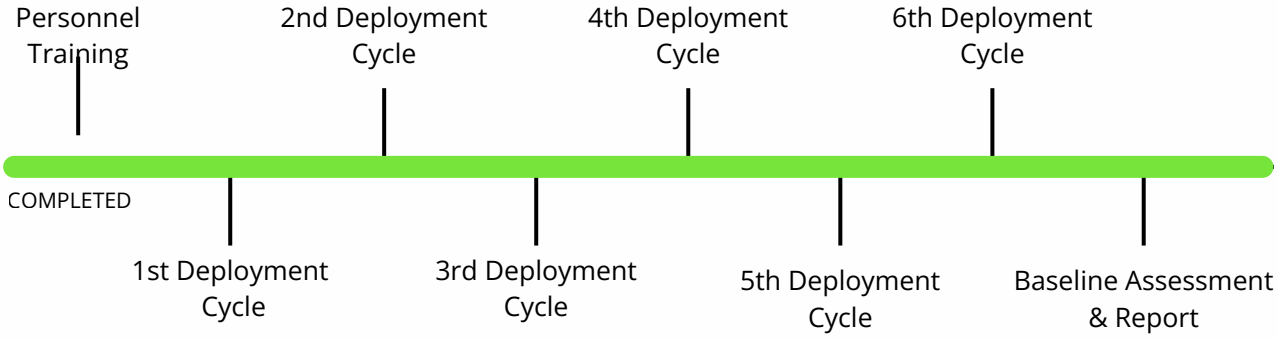
Enhance biodiversity by creating a favorable environment for associated flora and fauna.

- *Monitor and quantify improvements in carbon absorption and other environmental indicators.*
- ***Diversity of functions*** → *fruit* ***(Mango),*** *medicine & erosion control* ***(Arjun),*** *fast restoration &* ***soil improvement (Sonajhuri), long-term timber value (Sagwan****).*
- ***Livelihood security*** → *provides both short-term benefits* ***(fruit, fuelwood, fodder) and long-term value (timber, medicine, ecosystem services).***
- *Mango grows well in Dumka's soils, providing fruit, shade, and* ***pollinator support.***
- *Arjun thrives near water, prevents* ***soil erosion****, and has* ***medicinal value.***
- ***Sonajhuri is drought-resistant****, restores degraded land, and* ***enriches the soil.***
- ***Sagwan*** *(Teak) offers long-term timber value and* ***stabilizes forest canopies.***
- *Together, these species balance ecological restoration with community livelihood.*
- *They ensure* ***climate resilience, biodiversity support, and income security for Dumka.***

PROJECT DETAILS

Region of Interest: **Dumka**	**Project Area (ha):** **147.23**	**Number of Sites:** **70**
Total Recordings: **24K+**		**Project- Wide Unique Bird species:** **73**
Total Observations: **6761**		**Species Confidence threshold:** **90%**

PROJECT STATUS:



PROJECT GALLERY:

KEY FINDINGS:

6
Migratory birds

Tringa ochropus

Ficedula albicilla

Lanius cristatus

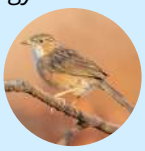
2
State High Priority Birds

Anthus hodgsoni

Psittacula eupatria

11
Endemic Birds

Vanellus malabaricus

Argya caudata


Dinopium benghalense

IUCN Red List

LC	VU	NT	EN	CR
Least Concerned 72	Vulnerable 0	Near Threatened **1 (Alexandrine Parakeet)**	Endangered 0	Critically Endangered 0

Keystone Species

There is no official government-published "keystone bird species of Jharkhand" list; instead, conservation literature and state habitat notes point to keystone-like guilds and representative species whose ecological roles structure Sal forests and wetlands.

Frugivores: key seed dispersers maintaining forest regeneration and plant diversity

Treron phoenicoptera

Psilopogon haemacephalus

Sturnia malabarica

Aerial insectivores and pest control: deliver sustained insect control across agroforestry and edge habitats

Dicrurus macrocercus

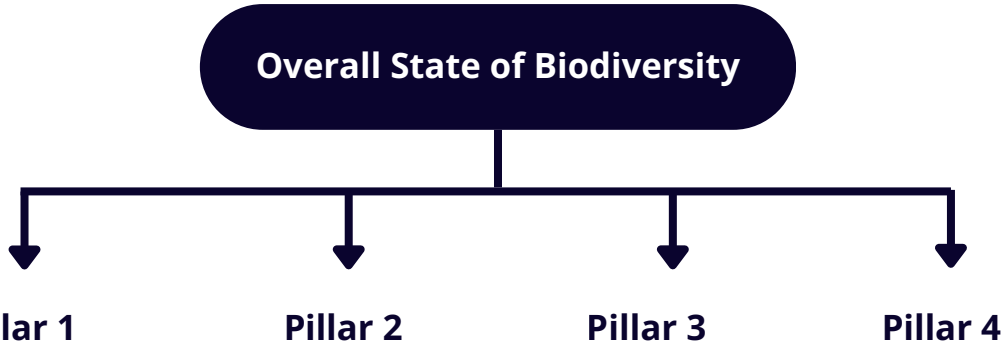
Merops orientalis

Cypsiurus balasiensis

Biodiversity Score:

Biodiversity is multi-faceted, variable in time and space, and often hard to observe directly. To give a clear, actionable picture we break biodiversity into four complementary pillars that together capture habitat, community structure, species status, and pressures — then combine those pillars into a single, transparent score you can use for monitoring, comparison and decision-making.

Our metric set is hybrid: pragmatic, field-measurable indicators from the **Singapore City Biodiversity Index (e.g., species counts, connectivity)** aligned with **TNFD/LEAP** disclosure principles, and supplemented by **global state and footprint measures (MSA, IUCN RLI)** for robustness. Acoustic disturbance metrics (NDSI and other soundscape indices) capture in-situ anthropogenic pressure mapped to the Threats pillar, making the overall score both locally actionable and globally comparable.



Ecosystem Condition	Species Assemblage	Species / Population Status	Threats & Pressures
The physical and functional support for life	The composition and diversity of the biological community	Direct measures of key species and population trends	The human and environmental drivers of decline

The scoring system

- **Metric → points:** Each metric is mapped to fixed points via evidence-based thresholds.
- **Pillar subtotals:** Metric points are summed into four pillar subtotals; each pillar has a defined maximum for comparability.
- **Project Score:** Pillar subtotals are combined by adding to get the final project score between 0–44 (0 = poorest; 44 = targets met).
- **Quality checks:** Year-to-year gains are only credited when statistically supported; missing/low-quality data are flagged and handled by transparent rescaling rules.
- **Transparent outputs:** Reports include the 0–44, four-pillar scores, and the metric-level point table for auditing and interpretation.

This method gives you a single easy-to-read score while keeping the ecological detail visible and auditable — suitable for monitoring change, prioritizing interventions, and reporting to partners.

Reference

https://www.cbd.int/article/2021-singapore-index

https://www.nparks.gov.sg/docs/default-source/resources/2021/handbook-singapore-index-cities-biodiversity.pdf

https://www.planvivo.org/Handlers/Download.ashx?IDMF=1c1aafdd-9c16-4b97-877b-391d96bcc038-

DUMKA, JHARKHAND BIODIVERSITY INDEX

INDICATORS

Core Pillars	Indicators	Points	Remarks
Pillar 1: Ecosystem Condition	1. Vegetation Structure- NDVI	2/4	The Ecosystem Condition is generally low because the plantation activities have just started.
	2.Habitat Health Index	2/4	
	3. CPLAND	1/4	Enhance habitat connectivity by planting trees
Pillar 2: Species Assemblage	4. Species Richness	1/4	Low species – habitat recovery needed.
	5. Species Diversity	3/4	Healthy baseline evenness in the avian community.
	6. Acoustic Health	2/4	Positive. A natural, biophony-dominated soundscape.
	7.Taxonomic Dissimilarity	2/4	Increase functional diversity
Pillar 3: Species Status	8. Mean Species Abundance (MSA)	1/4	Confirms restoration need; a key recovery metric.
	9. Conservation Value	3/4	Significant. Supports a Near Threatened species.
Pillar 4: Threats & Pressures	10. Land Anthropogenic Noise Intrusion (NDSI)	3/4	Excellent. A key asset attractive to wildlife.
	11. Human Intrusion (Proximity)	2/4	Identifies pressures; informs buffer zone strategy.
Pillar 1: Ecosystem Condition		5/12	
Pillar 2: Species Assemblage		8/16	
Pillar 3: Species / Population Status		4/8	
Pillar 4: Threats & Pressures		5/8	
Maximum Total:		**22/44**	

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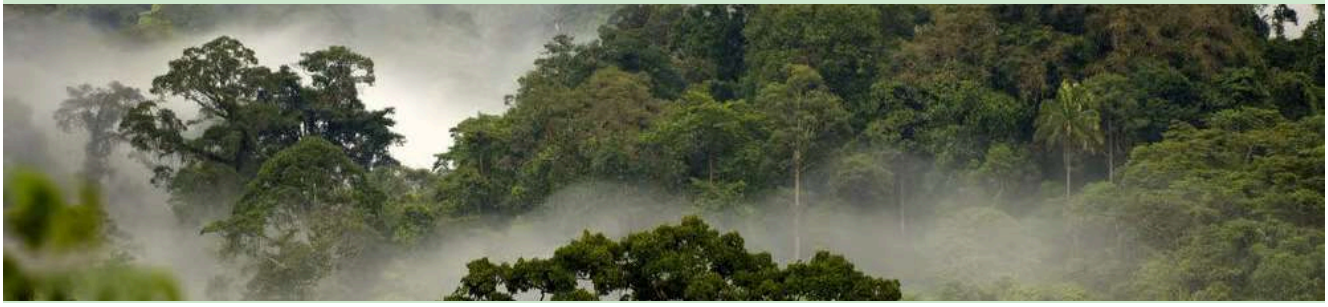
Pillar 1: Ecosystem Condition



Indicator 1: Vegetation Structure - NDVI

NDVI (Normalized Difference Vegetation Index) represents how green and vigorous the land's vegetation is at a given time; Areas with lush, lively plant growth show **higher NDVI values, while dry, degraded, or built-up areas show lower values**. The NDVI values presented are at the project level and **calculated for all 70 sites.**

$$NDVI = \frac{NIR - Red}{NIR + Red}$$

where *NIR* and *Red* represent surface reflectance in the near-infrared and red bands, respectively.

	Categorical values	Points
Excellent Vegetation	> 0.429	4
Good Vegetation	(0.402 - 0.429]	3
Moderate	(0.375 - 0.402]	2
Early Recovery	(0.336 - 0.375]	1
Severely Degraded	≤ 0.336	0

Project Average Value	Points Acquired
0.388	2

NDVI range (Mean ± SD): 0.388 ± 0.052

Recommendations:

*The average **NDVI (0.388) indicates moderate vegetation**, which is expected **at this baseline stage of plantation.** Scores are preliminary and expected to increase as plantations establish and vegetation matures.*

- ***Enrichment Planting:** Introduce native, multi-purpose species to enhance vegetation cover, prioritizing areas currently below moderate condition.*
- ***Enrich Soil:** Use compost and organic amendments during planting, especially in areas below the "Moderate" threshold, to improve water retention and nutrient availability.*
- ***Mandate Mulching:** Apply a 15 cm layer of organic mulch around all seedlings to conserve soil moisture and buffer against drought, a key climate risk.*
- ***Monitor Performance:** Track quarterly NDVI changes to assess the effectiveness of soil conservation measures and adapt management accordingly.*

Note:** These **baseline scores provide a starting point; as plantations develop, vegetation and associated NDVI scores are expected to improve**, reflecting progress in **habitat restoration.

Indicator 2: Habitat Health Index

The Habitat Health Index (HHI) represents how stable and vigorous a site's greenness is over time. It summarizes the level and stability of plant cover that wildlife depends on. Sites with consistently higher greenness tend to **support richer communities and better foraging and breeding conditions.**

HHI is derived from **Sentinel-2 NDVI**, which measures vegetation density and productivity.

$$HHI = \frac{z_5}{\sigma_{NDVI}}$$

where z_5 is the 5th percentile of NDVI values within the site and σ_{NDVI} is the NDVI standard deviation. An increase in HHI denotes habitat recovery or improved vegetation stability, whereas a decrease signals potential degradation.

	Categorical values	Points
Excellent Vegetation	> 1.947	4
Good Vegetation	(1.776 - 1.947]	3
Moderate	(1.603 - 1.776]	2
Early Recovery	(1.385 - 1.603]	1
Severely Degraded	≤ 1.385	0

Project Average Value	Points Acquired
1.707	**2**

HHI range (Mean ± SD): 1.707± 0.316

Recommendations:

*The Habitat Health Index **(1.707) indicates a fairly healthy habitat** with **moderate vegetation cover and biodiversity**. Vegetation is establishing well, but continued management is needed to strengthen ecosystem stability.*

- ***Establish Framework Species:*** *Deploy fast-growing **Acacia auriculiformis (Sonajhuri)** as a foundational matrix in low-HHI sites to rapidly stabilize green cover and improve the HHI score.*
- ***Develop Understory:*** *Introduce native grasses and shrubs to create a multi-layered habitat, further stabilizing the NDVI signal and protecting topsoil.*
- ***Adaptive Management:*** *Use annual HHI data to identify and remediate under performing sites with potential soil compaction or nutrient issues.*

With consistent management, the site can advance toward good habitat health (≥1.76) in the next monitoring cycle.

Indicator 3: CPLAND

CPLAND (Connected Patch of Low-Intensity Land) measures t**he connectivity of low-intensity habitats within the site,** reflecting the **landscape's capacity to sustain biodiversity.** Habitat patches are classified as low-intensity human use (e.g., semi-natural forests, agroforestry) or intensive use (e.g., crops, plantations). The index incorporates both the extent and spatial arrangement of low-intensity habitat patches. Higher values indicate better ecological connectivity and greater ecosystem resilience.

$$CPLAND = \left(\frac{\Sigma\,(a_i)}{A}\right) \times 100$$

where ai = area of each low-intensity habitat patch that is functionally connected to at least one other patch within a defined threshold distance and A is the total landscape area.

For the project, the **CPLAND value is 21.35%, indicating relatively low connectivity among low-intensity habitats**. **This low connectivity arises primarily because the project landscape is a mosaic of small agroforestry plots interspersed with intensive-use lands, such as crop fields and plantations.** The fragmentation limits functional linkage between habitat patches, restricting movement and dispersal for many species.

Since this represents **the baseline year, connectivity is expected to improve over time as plantations** mature and habitat patches expand, thereby **enhancing ecosystem resilience** and supporting greater biodiversity across the landscape.

	Categorical values	Points
Excellent connectivity	(80–100] %	4
Good connectivity	(60–80] %	3
Moderate connectivity	(40–60] %	2
Poor connectivity	(20–40] %	1
Very poor connectivity	[0–20] %	0

Project Average Value	Points Acquired
21.35%	1

Recommendations:

1.Expand Low-Intensity Habitats

- *Increase the extent of natural or semi-natural vegetation, such as native forests, agroforestry systems, and multi-species plantations.*
- *Prioritize areas currently dominated by intensive human use (e.g., monoculture crops or plantations) for gradual restoration.*
- *Include a mix of native trees, shrubs, and groundcover to create multi-layered habitats that support diverse species.*

2. Enhance Core Habitat Areas

- ***Create Core-Corridor Network:*** *Identify the five largest, highest-quality habitat patches as "core areas" and connect them with 10-meter-wide native vegetation corridors to measurably increase connectivity.*
- ***Prioritize Riparian Buffers:*** *Focus corridor creation along drainage lines, planting moisture-loving species like Terminalia arjuna to simultaneously enhance connectivity and mitigate erosion.*

3. ***Reduce Habitat Fragmentation***

- *Limit expansion of intensive land use within and around the project area.*
- *Avoid further clearing or degradation of existing low-intensity habitats.*
- *Implement sustainable land management practices in adjacent agricultural lands to reduce edge effects and maintain connectivity.*

4. ***Community Engagement and Capacity Building***

- *Involve local communities and landowners in planting and habitat restoration efforts.*
- *Provide training on sustainable land-use practices and the importance of habitat connectivity for biodiversity.*
- *Promote community-managed green belts around project sites to extend habitat and create buffer zones.*

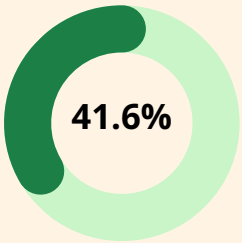
Consolidated Pillar Score:

Maximum Points:

12

Scored Points:

5



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Pillar 2: Species Assemblage

Indicator 4: Species Richness

Species richness refers to the total number of unique species recorded within a defined area. It is a fundamental measure of biodiversity, offering insight into the ecological complexity and health of an ecosystem. In this assessment, a total of **73 distinct species** were identified across the project sites.

For a broader context, secondary data available from citizen science platforms such as **eBird (Cornell Lab of Ornithology) show that 208 species** have been recorded so far in this region, indicating that the **field survey has captured ~37%** of the known diversity. This highlights both the richness already supported by the project nearby areas and the potential for further discoveries with continued monitoring.

Categorical values		Points
Very High Richness	> 80 %	4
Good Richness	(60 - 80] %	3
Moderate Richness	(40 - 60] %	2
Low Richness	(20 - 40] %	1
Severely low Richness	≤ 20 %	0

Project Value	Points Acquired
37 %	1

Recommendations:

- ***Target Keystone Guilds:*** *Dedicate 15% of planting area in low-richness sites to native, fruit-bearing shrubs (Ziziphus mauritiana, Carissa carandas) to support essential frugivore bird species.*
- ***Long-Term Monitoring & Research***
  - *Deploy bioacoustics, camera traps, and vegetation surveys for continuous tracking.*
  - *Use monitoring data to publish biodiversity indices annually, strengthening project credibility.*
- ***Guarantee Pollinator Support:*** *Establish dedicated "pollinator corridors" with high-nectar native plants to address the identified pollinator concern and support Mango crop productivity.*
- ***Build Vertical Structure:*** *Implement a three-tiered planting design (canopy, mid-story, shrub layer) to create diverse niches for nesting, foraging, and shelter.*

Summary:
Because landscape-level species richness is Low, the project should prioritize active habitat creation and threat reduction, with fast-acting connectivity interventions and guild-focused species support. A transparent monitoring program (acoustics + field) with coverage-standardized richness will track gains.

Indicator 5: Species Diversity

Species diversity is measured by considering both the number of species and their **relative abundances.** As ecosystems recover, species distributions typically become more balanced, shifting from dominance by a few species to a more even community structure.

Species diversity is calculated using the **Hill number (q = 1)** [Hill 1973], which corresponds to the exponential of the Shannon index.

$$Hill\,(q=1):^{1} D = exp\,(-\Sigma p_i ln\,(p_i))$$

where pi is the relative abundance of species i.

Diversity is first calculated separately for each target group (e.g., plants, birds, insects) and then summed to obtain a site-level diversity value. We are targeting only avian diversity as a proxy for biodiversity. Higher diversity values indicate **greater biodiversity and more resilient ecosystems.**

Categorical values		Points
Very High diversity	> 14.05	4
Good Diversity	(5.21 - 14.05]	3
Moderate diversity	(3.68 - 5.21]	2
Low diversity	(2.01- 3.68]	1
Severely low diversity	≤ 2.01	0

Site Average Value	Points Acquired
8.64	3

Recommendations:

- ***Habitat Enrichment:*** *Introduce native flowering and fruiting plants, as well as insect-host species, to attract pollinators, frugivores, and insectivorous birds.*
- ***Create Wetland Microhabitats:*** *Excavate small water harvesting ponds ('dobhas') at sites where Terminalia arjuna is planted. This stores water for drought mitigation and attracts a new guild of water-associated birds.*
- ***Manage Deadwood:*** *Retain existing dead trees (snags) and create habitat piles from pruned branches to support cavity-nesting and insectivorous birds.*
- ***Maintain Habitat Mosaics:*** *Intentionally preserve small patches of open grassland within larger sites to maintain habitat for edge species.*

[HILL 1973] *Hill, M.O., 1973. Diversity and evenness: a unifying notation and its consequences. Ecology, 54(2), pp.427-432.*

Indicator 6: Acoustic Health

Acoustic Health expresses how 'natural' and diverse a site's soundscape is. **High values indicate rich, varied biological activity (many species / varied calls) while low values indicate dominance by noise** or a few repetitive vocal sources.

Indices used to represent the overall Acoustic Health of the project:

- **Acoustic Complexity Index (ACI) [500Hz - 12kHz]:** Measures soundscape biodiversity through amplitude variability; Higher ACI → richer biotic activity.
- **Acoustic Diversity Index (ADI) [500Hz - 12kHz]:** Reflects distribution of sound across frequency bands; Higher ADI → greater acoustic diversity.
- **Acoustic Evenness Index (AEI) [500Hz - 12kHz]:** Indicates balance of sound energy across bands; Higher AEI → more even, balanced soundscape (diverse species).
- **Bioacoustics Index (BI) [2kHz - 8kHz]:** Integrates sound energy above a noise threshold across biologically relevant frequencies; higher BI → stronger biophonic activity.

Final metric combines **ACI, ADI, AEI, and BI** to provide an overall measure of the **site's ecological and acoustic health.**

$$Acoustic_{Health} = Median\,(Composite) - \lambda.\,IQR$$

where Composite is the weighted mean of scaled ACI, ADI, 1-AEI and BI; . Median of composite score is chosen as a robust central tendency and λ is a tunable penalty on stable variability i.e., IQR (Inter Quartile Range).

Combined acoustic indices can be a better indicator of biodiversity and ecosystem health than individually [Alcocer et al. 2022].

	Categorical values	Points
Excellent	> 0.509	4
Good	(0.501- 0.509]	3
Moderate	(0.494 - 0.501]	2
Poor	(0.448 - 0.494]	1
Very Poor	≤ 0.448	0

Site Average Value	Points Acquired
0.496	2

Acoustics Health (Mean ± SD): 0.496 ± 0.047

[Alcocer et al. 2022] Alcocer, I., Lima, H., Sugai, L. S. M., & Llusia, D. (2022). Acoustic indices as proxies for biodiversity: a meta-analysis https://doi.org/10.1111/brv.12890

Recommendations:

- ***Promote Vertical Stratification:*** *Implement a multi-layered planting design (canopy, mid-story, understory) to create distinct vocal niches, which is proven to increase the Acoustic Complexity Index (ACI).*
- ***Schedule Low-Noise Activities:*** *Restrict machinery use to midday hours, avoiding the critical dawn and dusk chorus periods of animal communication.*
- ***Expand Bioacoustic Monitoring:*** *Incorporate insect and amphibian call monitoring to create a more holistic measure of soundscape health.*

Goal: *Achieve a richer, more diverse soundscape, shifting toward* ***higher acoustic health (≥0.50),*** *indicating a balanced, biophony-dominated environment.*

Return to ES

Indicator 7: Taxonomic Dissimilarity

The Taxonomic dissimilarity among species reflects the diversity of evolutionary lineages within a community. It captures how taxonomically distinct the species are, indicating ecological complexity and resilience.

For this site—**an agricultural landscape where monitoring focuses on bird species—taxonomic dissimilarity is calculated using within-group variation (Δ*), which measures how different bird species are in their taxonomic relationships and relative abundances.**

Higher Δ* values indicate a greater range of bird lineages and a more balanced, resilient ecosystem, while lower values suggest reduced taxonomic diversity often associated with habitat disturbance or monoculture conditions.

Categorical values		Points
Excellent	> 0.958	4
Good	(0.876 - 0.958]	3
Moderate	(0.764 - 0.876]	2
Low	(0.649 - 0.764]	1
Severely low	≤ 0.649	0

Site Average Value	Points Acquired
0.824	2

Recommendations:

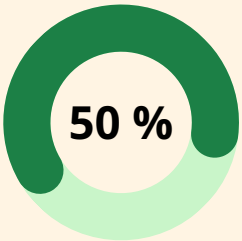
Enhance Taxonomic Representation:

- ***Promote habitat conditions*** *that support a wider range of taxonomic groups such as birds, insects, amphibians, and small mammals, by maintaining diverse microhabitats.*
- ***Establish Pollinator Baseline:*** *Implement a standardized pollinator monitoring protocol (e.g., pan traps) to address the "pollinator concern" and add a new taxonomic order (Hymenoptera) to the analysis.*
- ***Introduce Mammal Monitoring:*** *In Year 2, deploy camera traps at representative sites to add data on mammals, a critical functional group.*
- ***Integrate Amphibian Surveys:*** *Following the creation of 'dobhas', conduct annual auditory surveys for frog calls to add another sensitive indicator group.*

Consolidated Pillar Score:

Maximum Points: 16

Scored Points: 8



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Pillar 3: Individual Species / Population Status

Indicator 8: Mean Species Abundance (MSA)

The Mean Species Abundance (MSA) index measures the **average abundance of native species in a given area compared to their abundance in undisturbed ecosystems**. It reflects how **human pressures**—such as **land use change, fragmentation, pollution, and infrastructure—impact species populations** over time.

For this assessment, **the GLOBIO4 model** was used to estimate overall MSA values based on **land-use data and ecosystem condition.**

Categorical values		Points
Excellent/Near-Intact	> 0.7	4
Moderately Intact/Semi-natural	(0.5 - 0.7]	3
Heavily Modified/Fair	(0.3 - 0.5]	2
Poor/Degraded	(0.1 - 0.3]	1
Very Poor/Collapsed	≤ 0.1	0

Site Average Value	Points Acquired
0.285	**1**

MSA (Mean ± SD): 0.285 ± 0.053

Recommendations:

- ***Mandate Soil Bioremediation:*** *For all sites with "Poor/Degraded" MSA scores, mandate pre-planting soil remediation, including compost application and inoculation with native mycorrhizal fungi.*
- ***Control Invasive Species:*** *Implement a systematic program for the removal of invasive plants that outcompete native flora and suppress the MSA score.*
- ***Reduce Chemical Inputs:*** *Strictly limit the use of agrochemicals in and around project sites to protect soil microbial communities and non-target native species.*

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Indicator 9: IUCN RED List Index.

The conservation importance of the project area is quantified using a **Conservation Value** metric, which weights species based on their risk of extinction as categorized by the **International Union for Conservation of Nature (IUCN)** Red List. The metric is designed to highlight the presence of globally threatened and locally protected species as critical conservation assets.

Of the **73** species recorded, **72** are classified as **Least Concern (LC),** reflecting a healthy foundational community. However, the site's value is significantly boosted by the documentation of the **Alexandrine Parakeet** (*Psittacula eupatria*).

Classified as **Near Threatened (NT)**, this species is highly sensitive to habitat loss and trapping pressures. Its confirmed presence across monitoring sites serves as a **key ecological indicator**, validating the presence of good-quality habitat that is vital for the survival of vulnerable species.

Categorical values		Points
N/A (No species or highly invasive)	No value. No bird species or highly invasive species recorded	**0**
Least Concern	Foundational Conservation Value	**1**
Keystone/Specialist	Significant Functional value: Multiple (>1) keystone or habitat specialist species	**2**
Near Threatened	High Conservation Asset: Presence of Near Threatened (≥ 1) or State High Priority Birds	**3**
CR, EN, VU	Critical Conservation Asset: Presence of IUCN Vulnerable (VU), Endangered (EN), Critically Endangered (CR) or WLPA Schedule I species.	**4**

Conservation Value	Points Acquired
High Conservation Asset	**3**

Recommendations:

1. Protect Key Habitats & Species

- ***Protect Nesting Sites:*** *Identify and legally protect all mature trees with existing nesting cavities within a 50-meter "no-disturbance" buffer zone to safeguard the Near Threatened* ***Alexandrine Parakeet****.*
- ***Create Nesting Opportunities:*** *Install 10 artificial nest boxes designed for large parrots in sites where the parakeet has been recorded.*

2. Enhance Habitat Complexity

- *Introduce native vegetation layers (ground cover, shrubs, and canopy trees) to increase microhabitats and support diverse bird communities.*

3. ***Reduce Disturbances***

- *Limit pesticide use, uncontrolled grazing, and habitat clearing during the breeding season to minimize impacts on sensitive species.*

4. Control Invasive Species

- *Remove or manage invasive plants that degrade habitat quality and reduce food availability for native birds.*

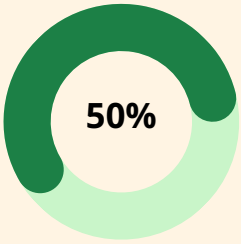
Consolidated Pillar Score:

Maximum Points:

8

Scored Points:

4



Return to ES

Pillar 4: Threats & Pressures

Indicator 10: Anthropogenic Sound (Normalized Difference Soundscape Index (NDSI))

This indicator translates the Normalized Difference Soundscape Index (NDSI) into a measure of biological dominance within the soundscape. **NDSI ranges from -1 (anthrophony-dominated) to +1 (biophony-dominated). Higher values indicate lower human noise and stronger biophonic activity, reflecting healthier, more natural soundscapes.** Use this indicator to identify areas or periods where biological sounds (birds, insects, amphibians) prevail over human-generated noise.

The NDSI average of **0.503** sits at the threshold between Moderate and Good Acoustic Health, reflecting a positive, biophony-dominated soundscape with manageable human noise intrusion.

	Categorical values	Points
Excellent	> 0.566	4
Good	(0.502 - 0.566]	3
Moderate	(0.484 - 0.502]	2
Poor	(0.267 - 0.484]	1
Very Poor	≤ 0.268	0

Site Average Value	Points Acquired
0.504	3

NSDI (Mean ± SD): 0.504 ± 0.151

Recommendations:

1. ***Create Vegetative Sound Buffers:*** *On site boundaries adjacent to roads or villages, plant dense, 5-meter-wide strips of fast-growing shrubs and trees (Sonajhuri) to absorb and deflect anthropogenic noise.*
2. ***Implement Low-Noise Management:*** *Restrict the use of loud machinery to midday hours (10 am - 4 pm) to avoid disrupting dawn and dusk animal communication periods.*
3. ***Zoning and Access Management****: Restrict access to high-value biodiversity areas during sensitive periods (e.g., breeding seasons).*
4. ***Community Awareness:*** *Educate local communities and workers about minimizing noise impacts on wildlife.*

Return to ES

Indicator 11: Human Intrusion (Proximity)

Human disturbance refers to the degree of pressure or **impact caused by human activities on ecosystems, habitats, and species.** It includes direct activities such as **deforestation, agriculture, grazing, fuelwood collection, infrastructure development, and settlements, as** well as indirect effects like **noise, pollution, and habitat fragmentation.**

It is usually expressed as a quantitative index **(0–1 or absolute values),** where higher values indicate greater ecological stress and lower values indicate better habitat integrity.

Value Categories		Points
Very low Human Disturbance	[0.0 – 0.2]	4 Points
Low Human Disturbance	(0.2 – 0.4]	3 Points
Moderate human Disturbance	(0.4 – 0.6]	2 Points
High Human Disturbance	(0.6 – 0.8]	1 Point
Very high Human Disturbance	> 0.8	0 Points

Site Average Value	Points Acquired
0.520	2 Points

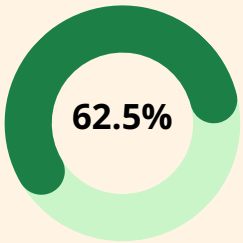
Recommendations:

1. ***Establish Community "Paryavaran Rakshaks":*** *Hire and train local community members as paid "Environment Guards" to monitor sites and conduct awareness campaigns, shifting from passive restriction to active stewardship.*
2. ***Develop Livelihood Alternatives:*** *Promote sustainable income generation (e.g., lac cultivation on Sonajhuri) to create a direct economic incentive for protecting project assets.*
3. ***Formalize Grazing Agreements:*** *Work with community leaders to establish designated grazing areas and rotational schedules to protect newly planted sites*

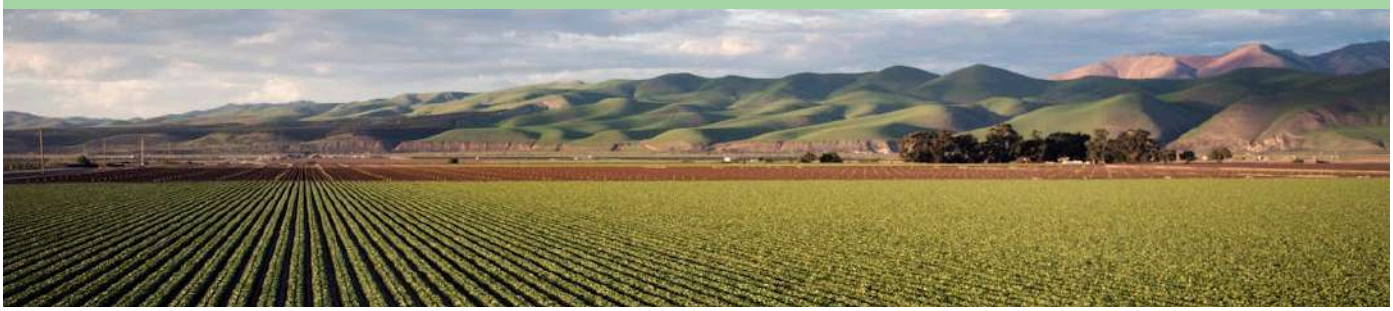
Consolidated Pillar Score:

Maximum Points: **8**

Scored Points: **5**



Climate-Related Risk and Opportunity Assessment for the Dumka Agroforestry



Portfolio (TCFD & TNFD Aligned Summary)

Introduction

This section provides a high-level, disclosure-ready summary of the **physical climate risks** and opportunities for the **70-site agroforestry** portfolio in Dumka, Jharkhand. **Based on downscaled climate projections to 2040**, the assessment is designed for integration into the Darukaa.earth's **Dumka Agroforestry: Year-0 Biodiversity Baseline & Monitoring Framework.**

Methodology

The analysis combines project-level biodiversity and landscape data with downscaled projections for three key hazards - **Extreme Heat, Drought, and Pluvial Flooding.** Using a **TCFD/CRSD-aligned framework**, it links quantitative c**limate risks with species-level physiological** responses to **identify operational vulnerabilities** and guide adaptive management strategies across the portfo**lio.**

Hazard	Risk & Impact up to 2040	Mitigation Strategy
Extreme Heat	**Significant Risk:** Up to 76 hot days/year; affects flowering and yields of Mango; increases water stress.	Use **heat-tolerant varieties, shade species** (Sonajhuri), and moisture conservation.
Drought	**High Risk**: 26 extreme drought days/year projected; threatens seedlings and moisture-dependent species (Arjun, Sagwan).	Apply **mulching**, **bunding**, **rainwater harvesting**; plant **drought-tolerant Sonajhuri**.
Pluvial Events & Erosion	**Moderate, Indirect Risk:** Higher rainfall intensity may cause erosion and waterlogging, harming Sagwan.	Use **cover crops**, **mulching**, **site drainage planning**; **Arjun** for riparian zones.

In-Depth Analysis of Climate Hazards and Species-Specific Vulnerabilities

Projected Climate Stressors for the Dumka Region

A detailed analysis of **climate projections** for the Dumka region reveals a pattern of increasing **environmental stress** characterized by **chronic heat,** more frequent and **intense droughts,** and **more extreme rainfall events.** These stressors, while distinct, are interconnected and create compound risks for the agroforestry project.

Climate Models / Scenarios	Primary Hazard	Key Results & Findings
SSP2–4.5, SSP3–7.0	**Extreme Heat**	• Significant Risk across most sites. • Days > threshold projected to rise to **75–76 days/year by 2040.** • Indicates chronic heat stress, **reducing flowering and fruit set in key species (e.g., Mango).** • Long-term productivity and economic viability at risk due to cumulative physiological stress.
RCP 8.5 (to 2040)	**Drought**	• High Risk with strong model agreement. • Extreme drought **(SPEI < –2.0)** days projected to rise from **< 1 to > 26 days/year.** • Threatens seedling establishment and **moisture-sensitive species (Arjun, Sagwan).** • Immediate adaptation needed despite possible reduction post-2050.
Long_SSP5–8.5	**Pluvial Events / Erosion**	• Direct flood risk is negligible, but 1-day **precipitation** is projected to **increase to ≈ 115 mm.** • On degraded, rocky terrain, it causes surface runoff, **erosion,** and **localized waterlogging.** • Highest vulnerability for **Sagwan (Teak), which requires well-drained soils.**

Return to ES

Species Vulnerability to Projected Climate Change

The FCF India portfolio includes four key species - **Mango, Arjun, Sonajhuri,** and **Sagwan,** chosen for fruit, timber, and ecological value. Their physiological traits create varying climate sensitivities: **Arjun is drought-prone, Sagwan is waterlogging-sensitive,** while **Sonajhuri shows strong resilience to multiple stressors.** Managing these interactions as an integrated system allows **Sonajhuri's resilience to buffer weaker species** and strengthen overall ecosystem stability. The matrix below summarizes **species-level vulnerabilities**.

Species	Extreme Heat	Drought	Pluvial Events (Erosion / Waterlogging	Overall Vulnerability
Mango (Mangifera indica)	High	Moderate	Low	High
Arjun (Terminalia arjuna)	Low	High	Moderate	High
Sonajhuri (Acacia auriculiformis)	Low	Low	Low	Low
Sagwan (Tectona grandis)	Moderate	High	High	High

The vulnerability matrix highlights each species' sensitivity to climate stressors in the Dumka agroforestry portfolio. **Mango, Arjun, and Sagwan face high overall risks due to heat, drought, or waterlogging, while Sonajhuri remains highly resilient, serving as a key species for enhancing site stability and supporting neighboring trees.** This matrix guides strategic species placement and climate-adaptive management.

Mango (*Mangifera indica*)

Risks:

Extreme Heat

Drought

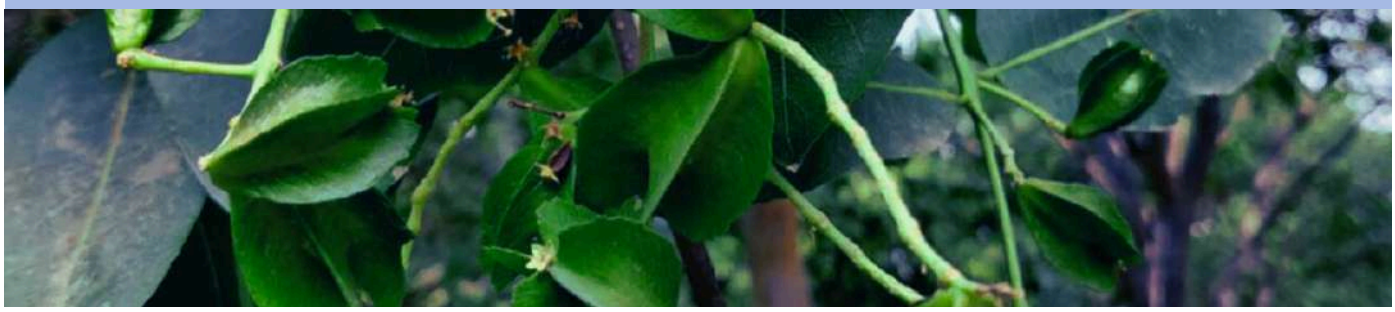
Impact Pathways and Scientific Validation

- ***Reproductive Failure:*** *High temperatures during flowering can dry out pollen and cause flowers or small fruits to drop, reducing the overall harvest.*
- ***Pollinator Disruption:*** *Hot weather stresses wild pollinators, making them less active, which further lowers fruit set.*
- ***Quality Decline:*** *Heat and drought can cause fruits to ripen poorly, become smaller, and lose juice, reducing market quality.*
- ***Higher Pest & Disease Risk:*** *Stress from heat and drought weakens the tree, making it more prone to pests and diseases.*

Mitigation Strategies

- ***Heat-Tolerant Varieties:*** *Plant Mango cultivars that flower later or are naturally tolerant to high temperatures, helping critical flowering avoid the peak summer heat.*
- ***Shade Trees:*** *Intercrop Mango with fast-growing species like Sonajhuri to provide shade, lower canopy temperatures, and reduce water loss.*
- ***Soil Moisture:*** *Apply thick mulch around the base and use basin irrigation to keep roots cool, retain soil moisture, and suppress weeds.*
- ***Tree Protection:*** *Use reflective clay sprays before heatwaves to reduce leaf temperature, and maintain a balanced nutrient program to strengthen tree resilience.*

Arjun (*Terminalia arjuna*)



Risks:

Drought

Waterlogging

Impact Pathways and Scientific Validation

- ***Drought Stress & Mortality:*** *Young seedlings and saplings are highly vulnerable to death from water shortage. Mature trees reduce growth by closing stomata, leading to leaf drop and branch dieback.*
- ***Waterlogging & Root Asphyxiation:*** *Arjun cannot tolerate prolonged soil saturation. Heavy rains on poorly drained soils can suffocate roots, cause root rot, and weaken the tree.*
- ***Root Destabilization:*** *Shallow root systems on slopes can be exposed to erosion during intense rainfall, reducing stability and nutrient uptake.*

Mitigation Strategies

- ***Site-Specific Planting (Hydro-zoning):*** *Plant in moist micro-sites like drainage lines or near ponds, ensuring good sub-surface drainage to prevent waterlogging.*
- ***Water Harvesting & Management:*** *Build contour trenches, swales, or check dams upslope to slow runoff, reduce erosion, and recharge soil water for use during drought.*
- ***Soil Preparation & Mulching:*** *Enrich planting pits with compost and other organic matter to improve water retention. Apply a thick layer of mulch to conserve moisture and protect roots during dry periods.*

Sonajhuri (*Acacia auriculiformis*)

Risks:

Extreme Heat

Drought

Seasonal Waterlogging

Impact Pathways and Scientific Validation

- ***Monetizing Resilience through Lac Cultivation:*** *Planting Sonajhuri as a nurse crop buffers against extreme heat and drought while serving as a host for lac insects, creating a short-to-medium-term income stream without extra land or water.*
- ***Risk Mitigation:*** *The multi-layered canopy of Sonajhuri and Mango trees protects lac from non-seasonal and intense rainfall, safeguarding production.*
- ***Co-Benefits:*** *This approach diversifies community livelihoods, generates early revenue before timber and fruit yields mature, and enhances biodiversity by adding a new ecological function—demonstrating a climate-smart, nature-based solution that directly links resilience to economic opportunity.*

Mitigation Strategies

- ***Nurse Crop:*** *Plant Sonajhuri around or between sensitive species to provide shade, wind protection, and better soil conditions.*
- ***Mulch from Pruning:*** *Regularly prune Sonajhuri and use the cut leaves/stems as mulch around other trees to protect roots from heat and drought.*
- ***Erosion Control:*** *On sloped sites, plant densely along contours to stabilize soil and reduce runoff during heavy rain.*

Sagwan (*Tectona grandis* / Teak)

Risks:

Extreme Heat

Drought

Waterlogging

Soil Erosion

Impact Pathways and Scientific Validation

- ***Seedling Death:*** *Young trees may die during dry periods.*
- ***Root Damage:*** *Waterlogged soil can suffocate roots, causing rot.*
- ***Soil Erosion:*** *Heavy rain on slopes can wash away soil, weakening root stability.*

Mitigation Strategies

- ***Choose the Right Site:*** *Plant only on deep, fertile, well-drained soils; avoid low-lying or clay-heavy areas.*
- ***Water Support:*** *Provide temporary irrigation for seedlings during dry spells.*
- ***Improve Drainage:*** *Build raised mounds or beds to keep roots above water.*
- ***Intercropping:*** *Plant with Sonajhuri or other nurse species to provide shade, reduce soil erosion, and protect against drought.*

An Integrated Strategy for Agroecosystem Resilience

Climate-Smart Planting and Portfolio Management

Building a **resilient agroforestry** system requires moving beyond monocultures to a diversified, climate-smart mosaic. Species should be strategically mixed to match site conditions and **reduce climate risks.**

Species	Planting Strategy	Soil & Water Conservation	Monitoring / Adaptive Management
Mango	• Use drought-tolerant or late-flowering varieties. • Intercrop with Sonajhuri for shade and better microclimate.	• Mulch around trees to retain moisture. • Apply basin irrigation to keep roots hydrated.	• Track flowering and fruiting times against temperature. • Monitor seedling survival during drought periods.
Arjun	• Plant in moisture-retentive valley bottoms or near water bodies. • Avoid areas prone to waterlogging.	• Mulch and enrich planting pits with compost to retain moisture. • Construct swales or small ponds to store runoff.	• Monitor seedling survival and soil moisture in riverine sites. • Track growth after rainfall events.
Sonajhuri	• Use as a nurse or framework crop on dry or degraded slopes. • Plant around sensitive species to provide shade and wind protection.	• Apply pruned biomass as mulch (“chop-and-drop”). • Plant along contour lines to prevent erosion.	• Monitor soil fertility, canopy cover, and moisture retention. • Track its protective effects on neighboring species.
Sagwan	• Plant on well-drained, fertile, elevated mounds. • Intercrop with Sonajhuri for protection.	• Mulch roots and provide temporary irrigation during dry spells. • Build raised beds on marginally drained soils.	• Track seedling survival, especially during drought. • Monitor soil saturation and erosion impacts.

Site level details

Restoration Site Baseline:

Satellite Imagery

A sample of **12 sites** was selected to assess the biodiversity present across different landscapes. These sites were chosen to represent a range of habitat types, land-use patterns, and ecological conditions. By studying them, the goal is to evaluate variations in **species richness, habitat quality, and connectivity.**

The findings from these **11 sites (encircled site is pending) provide a baseline understanding of how biodiversity responds to habitat fragmentation and environmental pressures.** This information will support the development of effective monitoring methods, guide **restoration planning,** and serve as a reference point for scaling assessments to larger landscapes.

Methods

Data Collections

At each site, bioacoustic recorders were deployed in **5-day monitoring cycles** to capture continuous soundscapes. These recordings enable detection and identification of bird species through their vocalizations, providing a non-invasive, standardized method **to track species richness, diversity, and habitat use.** Repeated 5-day cycles ensure temporal coverage and reduce sampling bias, allowing reliable comparisons across sites.

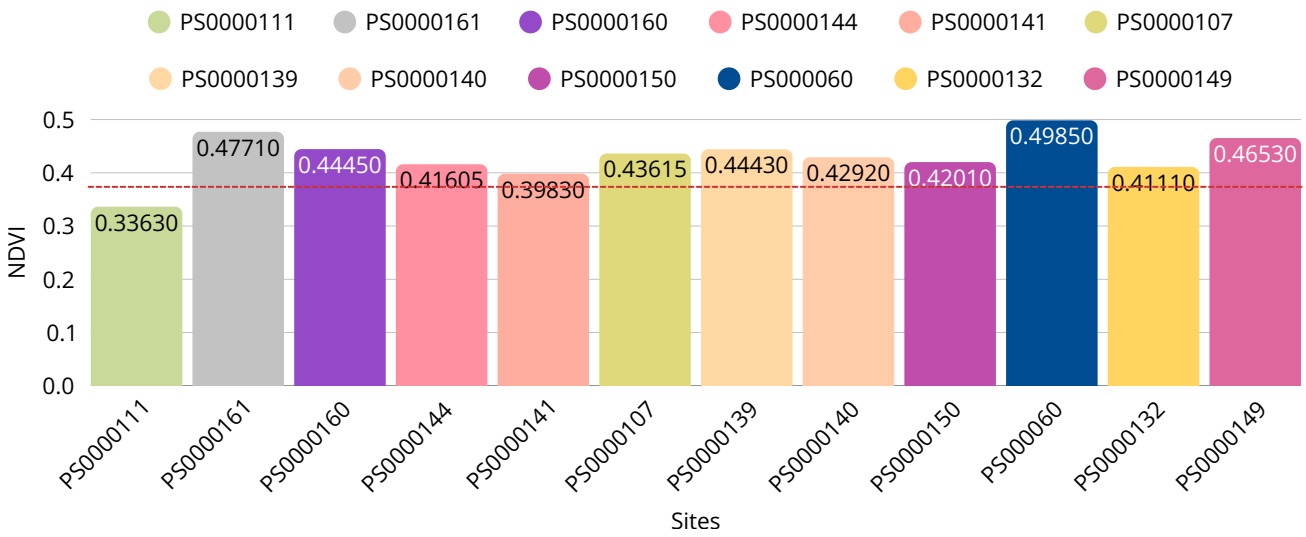
Locations ID	Latitute	Longitude	Deployment Start Date	Deployment End Date	Season
PS0000111	24.208041	87.319575	15/7/2025	19/7/2025	Monsoon
PS0000161	24.211955	87.328842	15/7/2025	19/7/2025	Monsoon
PS0000160	24.222517	87.307733	23/7/2025	27/7/2025	Monsoon
PS000144	24.21843	87.243247	23/7/2025	23/7/2025	Monsoon
PS000141	24.210457	87.363224	06/08/2025	11/08/2025	Monsoon
PS000107	24.192362	87.387955	20/8/2025	25/8/2025	Monsoon
PS000139	24.181184	87.399827	20/8/2025	25/08/2025	Monsoon
PS0000140	24.170646	87.401098	06/09/2025	11/09/2025	Autumn
PS0000150	24.224905	87.43124	06/09/2025	11/09/2025	Autumn
PS000060	24.207825	87.451583	20/09/2025	25/09/2025	Autumn
PS0000132	24.55711	87.31957	30/9/2025	05/10/2025	Autumn

*Monitoring started in **July and concluded in October**, covering the core monsoon period to capture peak seasonal bird activity.*

Pillar 1: Ecosystem Condition of Site Level

NDVI (Year-0 Baseline)

Normalized Difference Vegetative Index (NDVI) represents how green and vigorous the land's vegetation is at a given time. Figure (below) shows site-level NDVI (mean) values for each of the chosen 12 sites. Dashed red line shows the project average NDVI value **(0.388)**.



Uses of NDVI

- **Monitor vegetation health:** Detect stress from drought, disease, or human impact.
- **Estimate biomass and productivity:** Used in ecological and agricultural studies.
- **Detect land cover changes:** Track deforestation, reforestation, or habitat degradation.
- **Support conservation planning:** Identify priority areas for restoration or protection.

Interpretation:
NDVI values across the monitored sites ***(ranging from 0.33 to 0.50) indicate an overall trend of moderate to good vegetation health****. Most areas fall within the* ***recovery to healthy categories, showing positive vegetation cover and ecosystem activity.*** *Sites with NDVI above* ***0.45 exhibit good vegetation*** *vigor, while a* ***few around 0.33–0.40 represent early recovery or mild degradation zones.***

This NDVI assessment has been conducted across multiple project sites, enabling comparative insights into vegetation condition and restoration progress at a landscape scale. The class-wise distribution across all sites shows that:

- ***30–35% of sites fall in the Good vegetation class (>0.45)***
- ***45–50% are in Moderate to Early Recovery (0.38–0.45)***
- ***15–20% show Low or Degraded vegetation (<0.38)***

Note: *Site-level categorization has been done based on Leave-one-out cross validation.*

Site Code	NDVI Value	Category	Interpretation
PS0000111	0.33630	**Severely Degraded**	Very low vegetation cover; indicates degradation or stressed area.
PS0000161	0.47710	**Good Vegetation**	Dense and healthy vegetation with strong greenness.
PS0000160	0.44450	**Good Vegetation**	Well-developed vegetation cover and active photosynthesis.
PS0000144	0.41605	**Good Vegetation**	Lush vegetation and stable plant growth.
PS0000141	0.39830	**Moderate**	Moderately vegetated area showing partial recovery.
PS0000107	0.43615	**Good Vegetation**	High vegetation density indicating good ecological condition.
PS0000139	0.44430	**Good Vegetation**	Healthy vegetation structure and canopy cover.
PS0000140	0.42920	**Excellent Vegetation**	Strong vegetation vigor and canopy cover.
PS0000150	0.42010	**Excellent Vegetation**	Productive vegetation and positive habitat condition.
PS000060	0.49850	**Good Vegetation**	Very good vegetation growth and stable green cover.
PS0000132	0.41110	**Excellent Vegetation**	Healthy and productive vegetation layer.
PS0000149	0.4653	**Excellent Vegetation**	Strong vegetation vigor and canopy cover.

Recommendations:

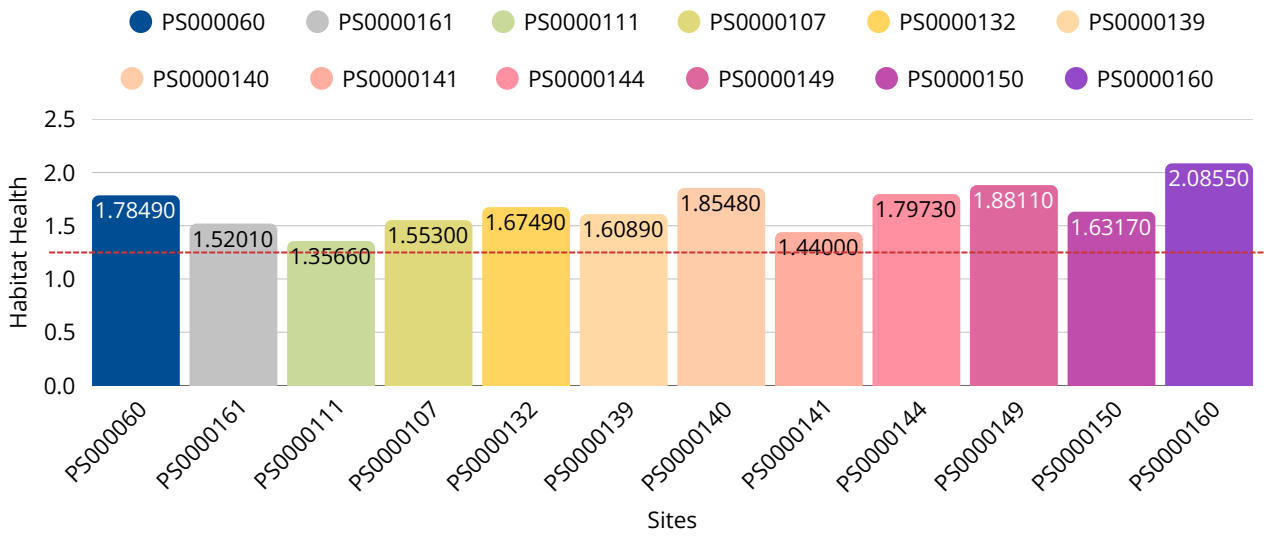
- ***High Priority (NDVI < 0.375; e.g., PS0000111)****: Implement intensive soil and water conservation, including contour trenches and heavy mulching, as a prerequisite for planting.*
- ***Moderate Priority (e.g, PS0000141)****: Focus on enrichment planting with a diverse mix of native species to accelerate canopy closure.*
- ***Low Priority (High NDVI sites; e.g., PS0000161)****: Prioritize protection from grazing and human disturbance to preserve existing healthy vegetation.*

Habitat Health:

The Habitat Health Index (often calculated using NDVI-based metrics) reflects the structural and functional quality of vegetation cover, which serves as a proxy for ecosystem productivity, habitat complexity, and resilience. In this case, the index was derived using:

- **5th percentile NDVI (Z5) – indicating vegetation health at its weakest points**
- **Standard deviation of NDVI (σ) – reflecting spatial variability in greenness**
- **Formula: P4 = Z5 / σ**

A higher HHI value generally indicates healthier, more stable habitats, while lower values suggest degradation, fragmentation, or stress.



Interpretation:

1. *The **Habitat Health Index** reflects the overall condition of vegetation and ecosystem stability across sites. Dashed red line shows the project average HHI value (1.707).*
2. ***High habitat health sites (e.g., PS0000140, PS0000149, PS0000160):** These areas have strong vegetation cover and biodiversity. They should be prioritized for **protection and long-term conservation to preserve ecosystem functions.***
3. ***Moderate habitat health sites (e.g., PS0000132, PS0000150):** These sites are functioning but remain **vulnerable to stress or degradation.** They require sustained management and selective restoration to enhance stability.*
4. ***Low habitat health sites (e.g., PS0000111, PS0000141):** These areas show signs of degradation and weak vegetation cover, likely due to human or environmental pressures. They should be prioritized for active restoration through native species planting, soil improvement, and connectivity enhancement.*
5. ***Priority Sites: PS0000111** and **PS0000141** have the **lowest habitat health values** and should be targeted first for restoration efforts.*

Overall Observation:
*Habitat health varies across the landscape, highlighting the need for site-specific strategies. Well-preserved habitats should be maintained as biodiversity strongholds, while degraded areas should be restored to **strengthen ecosystem resilience and connectivity.***

Site Code	Habitat Health Value	Interpretation	Score (0–4)
PS000060	1.78490	**Good habitat health, strong vegetation and biodiversity presence**	3
PS0000161	1.52010	**Fairly healthy habitat, moderate vegetation cover and biodiversity**	2
PS0000111	1.35660	**Early recovery stage, patchy vegetation and weak ecological support**	1
PS0000107	1.55300	**Fairly healthy habitat, moderate vegetation cover and biodiversity**	2
PS0000132	1.67490	**Fairly healthy habitat, moderate vegetation cover and biodiversity**	2
PS0000139	1.60890	**Fairly healthy habitat, moderate vegetation cover and biodiversity**	2
PS0000140	1.85480	**Good habitat health, strong vegetation and biodiversity presence**	3
PS0000141	1.44000	**Early recovery stage, patchy vegetation and weak ecological support**	1
PS0000144	1.79730	**Good habitat health, strong vegetation and biodiversity presence**	3
PS0000149	1.88110	**Good habitat health, strong vegetation and biodiversity presence**	3
PS0000150	1.63170	**Fairly healthy habitat, moderate vegetation cover and biodiversity**	2
PS0000160	2.08550	**Excellent habitat health, dense vegetation and high species richness**	4

Recommendations for Habitat Restoration.

. ***Site-specific Prioritization***

Degraded / Low Health Sites (PS0000111, PS0000161, PS0000141):

- *Prioritize soil enrichment (compost, mulching, biofertilizers).*
- *Introduce fast-growing pioneer species (Sonajhuri,) to stabilize soil and provide shade.*
- *Regular weeding to reduce competition with invasive species.*

Moderate / Fairly Healthy Sites (PS0000107, PS0000132, PS0000139, PS0000150):

- *Maintain a diverse species mix, including Mango and Sagwan, to develop long-term canopy cover.*

Good / High Health Sites (PS000060, PS0000140, PS0000144, PS0000149):

- *Enrichment planting of fruiting and flowering trees (Mango, Arjun) to support pollinators and fauna.*
- *Encourage natural regeneration by protecting existing seedlings.*

Excellent Habitat Health Sites (PS0000160):

- *Maintain habitat condition with minimal disturbance.*
- *Support natural regeneration processes.*

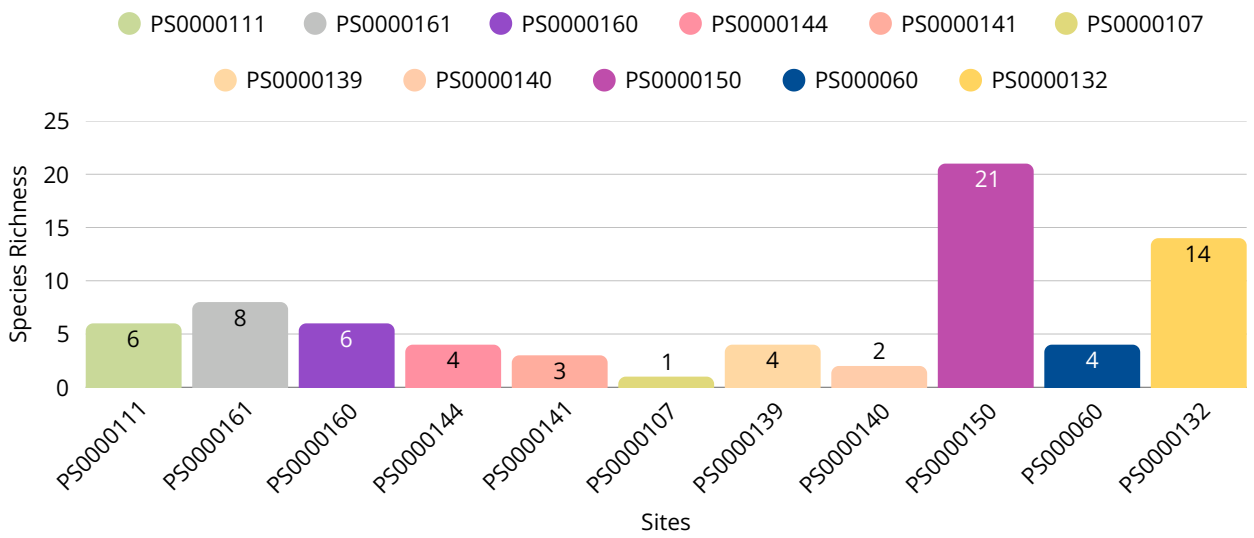
Pillar 2: Species Assemblage of Site Level

Species Richness:

Species richness refers to the total number of unique species recorded within a defined area. It is a fundamental measure of biodiversity, offering insight into the ecological complexity and health of an ecosystem. A richness value of **73 species indicates low richness as compared to species reported across the ecoregion**

Importance of Species Richness:

- **Ecosystem Health:** A higher number of species reflects a balanced and functioning ecosystem.
- **Resilience to Disturbance**: Diverse species composition helps ecosystems recover from stress, such as climate change or habitat loss.
- **Ecological Functioning:** Supports vital processes like pollination, nutrient cycling, and soil regeneration.
- **Habitat Quality Indicator:** High richness signifies suitable habitat conditions and resource availability.



Recommendation

To enhance species richness across the sites, the following actions are being implemented:

- ***Very Low Richness Sites (<5 species; e.g., PS0000107, PS0000140):*** *Focus on foundational habitat creation: establish a canopy, create habitat piles (deadwood), and excavate small water pits ('dobhas').*
- ***Low Richness Sites (6-8 species; e.g., PS0000111, PS0000160):*** *Enhance structural complexity by introducing a diverse understory of native fruiting and flowering shrubs.*
- ***Good Richness Sites (>14 species; e.g., PS0000150, PS0000132): P****rioritize protection. Maintain existing vegetation, protect mature trees, and establish buffer zones to shield from disturbance.*

Site Code	Species Richness	Interpretation	Recommendation
PS0000111	6	**Low richness**	Continue habitat enhancement through native tree planting; mango, arjun, sonajhuri, and sagwan will help attract more bird species.
PS0000161	8	**Low richness**	Maintain and expand native vegetation cover; increasing canopy diversity will support higher species richness.
PS0000160	6	**Low richness**	Strengthen habitat structure with mixed native species plantations to improve foraging and nesting opportunities.
PS0000144	4	**Very low richness**	Focus on microhabitat creation and understorey vegetation; the ongoing tree planting will gradually enhance bird diversity.
PS0000141	3	**Very low richness**	Improve water and food availability through continued plantation and protection of regenerating areas.
PS0000107	1	**Very low richness**	Enhance vegetation density and connectivity using mango, arjun, sonajhuri, and sagwan; this will aid species colonization.
PS0000139	4	**Very low richness**	Continue native tree planting and minimize disturbance to encourage more avian activity.
PS0000140	2	**Very low richness**	Strengthen vegetation recovery and maintain plantation efforts for gradual species increase.
PS0000150	21	**Good richness**	Maintain current vegetation management; protect mature trees and nesting habitats.
PS000060	4	**Very low richness**	Continue habitat enrichment through diverse native plantations to boost local bird presence.
PS0000132	14	**Good richness**	Maintain habitat quality and enhance buffer vegetation to sustain and attract more species.

Summary:

*The trees FCF is planting **(Mango, Arjun, Sonajhuri, Sagwan)** are strong choices because they **balance ecological restoration (soil, water, biodiversity) with community benefits (fruit, timber, livelihoods).** To maximize impact, they should be mixed with native flowering/fruiting species and structured into layered habitats. This approach will improve biodiversity, carbon sequestration, pollinator health, and resilience — while still supporting local people.*

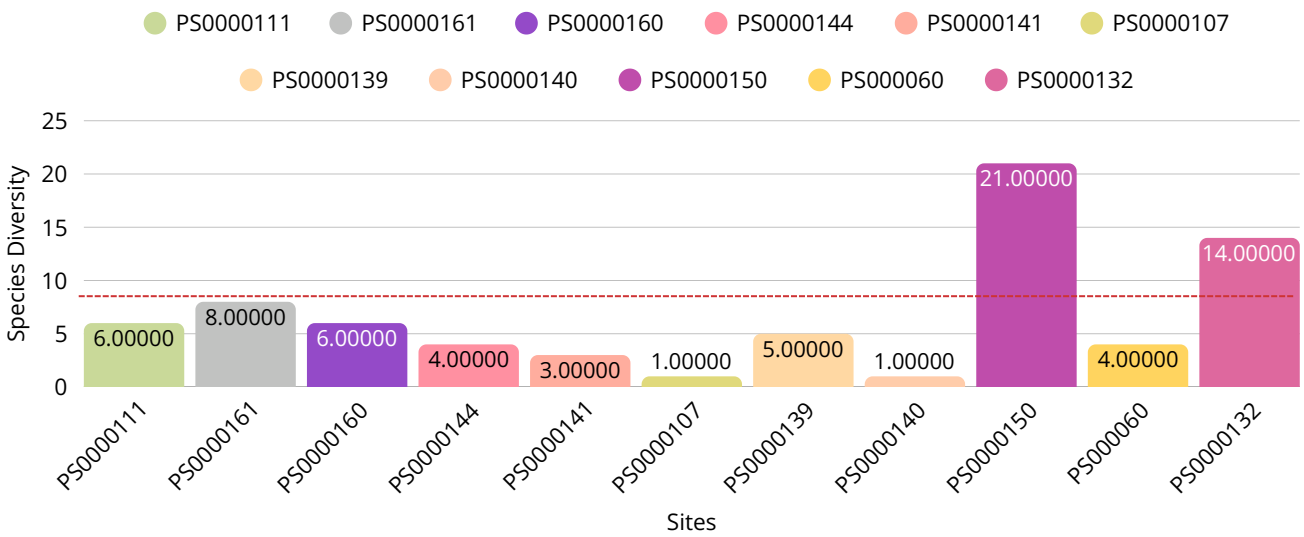
Species Diversity:

Species diversity refers to the variety and abundance of different species within a habitat or ecosystem. It combines both species richness **(the number of species present) and species evenness (how evenly individuals are distributed among those species).**

Importance of Species Diversity:

- Enhances ecosystem stability and **resilience to disturbances.**
- Maintains **ecological balanc**e and supports key functions like pollination and nutrient cycling.
- Improves **productivity** and **climate adaptability.**
- Serves as a key i**ndicator of habitat health** and **restoration success.**

Promoting species diversity is essential for sustaining ecosystem recovery and long-term ecological resilience.



Recommendation

To enhance species diversity across the sites, the following key actions are being implemented:

- ***Native Tree Plantation:*** *Continued planting of mango, arjun, sonajhuri, and sagwan to enrich habitat structure and food resources.*
- ***Habitat Restoration:*** *Strengthening vegetation layering and understorey growth to* ***create suitable nesting and foraging zones.***
- ***Water Resource Development:*** *Establishing small water bodies or moist patches to attract more bird species.*
- ***Disturbance Control****: Reducing grazing and human interference to allow natural regeneration.*
- ***Biodiversity Monitoring****: Using bioacoustic and field surveys to track changes in bird diversity and guide adaptive restoration.*

Site Code	Species Richness	Interpretation	Recommendation
PS0000111	6	**Low diversity**	Continue native tree planting and improve vegetation layering.
PS0000161	8	**Low diversity**	Add more fruiting and flowering trees to attract diverse birds.
PS0000160	6	**Low diversity**	Strengthen understorey and maintain habitat protection.
PS0000144	4	**Very low diversity**	Increase native cover and reduce disturbance.
PS0000141	3	**Very low diversity**	Improve water and food availability through plantations.
PS0000107	1	**Extremely low diversity**	Focus on rapid vegetation recovery and site protection.
PS0000139	5	**Very low diversity**	Enrich with mixed native species and maintain continuity.
PS0000140	1	**Extremely low diversity**	Restore degraded areas and prevent grazing pressure.
PS0000150	21	**Good diversity**	Maintain vegetation quality and protect nesting trees.
PS000060	4	**Very low diversity**	Expand plantation density and monitor species response.
PS0000132	14	**Moderate diversity**	Continue current restoration and enhance habitat buffers.

Summary:

The assessment of bird species across the project sites shows that most locations currently ***have low to very low species diversity, indicating early ecological recovery stages.*** *However, sites* ***like PS0000150 and PS0000132 demonstrate good diversity,*** *reflecting healthier habitat conditions.*

Avian Taxonomy

Taxonomy is the scientific system of classifying and naming living organisms based on their shared characteristics and evolutionary relationships. It helps organize biodiversity into hierarchical levels such as Kingdom, Phylum, Class, Order, Family, Genus, and Species. Below is the avian taxonomy for the detected bird species in the region.

scientific_name	matched_name	order	family	genus	species
Prinia hodgsonii	Prinia hodgsonii Blyth, 1844	Passeriformes	Cisticolidae	Prinia	Prinia hodgsonii
Otus bakkamoena	Otus bakkamoena Pennant, 1769	Strigiformes	Strigidae	Otus	Otus bakkamoena
Vanellus malabaricus	Vanellus malabaricus (Boddaert, 1783)	Charadriiformes	Charadriidae	Vanellus	Vanellus malabaricus
Argya striata	Argya striata (Dumont, 1823)	Passeriformes	Leiothrichidae	Turdoides	Argya striata
Pitta brachyura	Pitta brachyura (Linnaeus, 1766)	Passeriformes	Pittidae	Pitta	Pitta brachyura
Athene brama	Athene brama (Temminck, 1821)	Strigiformes	Strigidae	Athene	Athene brama
Psittacula eupatria	Psittacula eupatria (Linnaeus, 1766)	Psittaciformes	Psittacidae	Psittacula	Psittacula eupatria
Merops orientalis	Merops orientalis Latham, 1801	Coraciiformes	Meropidae	Merops	Merops orientalis
Ortygornis pondicerianus	Ortygornis pondicerianus (J.F.Gmelin, 1789)	Galliformes	Phasianidae	Ortygornis	Ortygornis pondicerianus
Lonchura punctulata	Lonchura punctulata (Linnaeus, 1758)	Passeriformes	Estrildidae	Lonchura	Lonchura punctulata
Dendrocitta vagabunda	Dendrocitta vagabunda (Latham, 1790)	Passeriformes	Corvidae	Dendrocitta	Dendrocitta vagabunda
Aegithina tiphia	Aegithina tiphia (Linnaeus, 1758)	Passeriformes	Aegithinidae	Aegithina	Aegithina tiphia
Streptopelia chinensis	Streptopelia chinensis (Scopoli, 1786)	Columbiformes	Columbidae	Spilopelia	Streptopelia chinensis
Coracina macei	Coracina macei (Lesson, 1831)	Passeriformes	Campephagidae	Coracina	Coracina macei
Prinia inornata	Prinia inornata Sykes, 1832	Passeriformes	Cisticolidae	Prinia	Prinia inornata
Nycticorax nycticorax	Nycticorax nycticorax (Linnaeus, 1758)	Pelecaniformes	Ardeidae	Nycticorax	Nycticorax nycticorax
Orthotomus sutorius	Orthotomus sutorius (Pennant, 1769)	Passeriformes	Cisticolidae	Orthotomus	Orthotomus sutorius
Cacomantis passerinus	Cacomantis passerinus (Vahl, 1797)	Cuculiformes	Cuculidae	Cacomantis	Cacomantis passerinus
Chrysomma sinense	Chrysomma sinense (Gmelin, 1789)	Passeriformes	Sylviidae	Chrysomma	Chrysomma sinense
Pelargopsis capensis	Pelargopsis capensis (Linnaeus, 1766)	Coraciiformes	Alcedinidae	Pelargopsis	Pelargopsis capensis
Psittacula cyanocephala	Psittacula cyanocephala (Linnaeus, 1766)	Psittaciformes	Psittacidae	Psittacula	Psittacula cyanocephala
Gallus gallus	Gallus gallus (Linnaeus, 1758)	Galliformes	Phasianidae	Gallus	Gallus gallus
Terpsiphone paradisi	Terpsiphone paradisi (Linnaeus, 1758)	Passeriformes	Monarchidae	Terpsiphone	Terpsiphone paradisi
Hierococcyx varius	Hierococcyx varius (Vahl, 1797)	Cuculiformes	Cuculidae	Cuculus	Hierococcyx varius
Oriolus xanthornus	Oriolus xanthornus (Linnaeus, 1758)	Passeriformes	Oriolidae	Oriolus	Oriolus xanthornus
Gracupica contra	Gracupica contra (Linnaeus, 1758)	Passeriformes	Sturnidae	Gracupica	Gracupica contra
Phylloscopus trochiloides	Phylloscopus trochiloides (Sundevall, 1837)	Passeriformes	Phylloscopidae	Phylloscopus	Phylloscopus trochiloides
Oriolus chinensis	Oriolus chinensis Linnaeus, 1766	Passeriformes	Oriolidae	Oriolus	Oriolus chinensis
Cypsiurus balasiensis	Cypsiurus balasiensis (J.E.Gray, 1829)	Apodiformes	Apodidae	Cypsiurus	Cypsiurus balasiensis
Halcyon smyrnensis	Halcyon smyrnensis (Linnaeus, 1758)	Coraciiformes	Alcedinidae	Halcyon	Halcyon smyrnensis
Cuculus micropterus	Cuculus micropterus Gould, 1838	Cuculiformes	Cuculidae	Cuculus	Cuculus micropterus
Caprimulgus asiaticus	Caprimulgus asiaticus Latham, 1790	Caprimulgiformes	Caprimulgidae	Caprimulgus	Caprimulgus asiaticus
Dinopium benghalense	Dinopium benghalense (Linnaeus, 1758)	Piciformes	Picidae	Dinopium	Dinopium benghalense
Lanius schach	Lanius schach Linnaeus, 1758	Passeriformes	Laniidae	Lanius	Lanius schach
Sturnia pagodarum	Sturnia pagodarum (Gmelin, 1789)	Passeriformes	Sturnidae	Sturnia	Sturnia pagodarum
Centropus sinensis	Centropus sinensis (Stephens, 1815)	Cuculiformes	Cuculidae	Centropus	Centropus sinensis
Hypothymis azurea	Hypothymis azurea (Boddaert, 1783)	Passeriformes	Monarchidae	Hypothymis	Hypothymis azurea
Centropus bengalensis	Centropus bengalensis (Gmelin, 1788)	Cuculiformes	Cuculidae	Centropus	Centropus bengalensis
Acridotheres tristis	Acridotheres tristis (Linnaeus, 1766)	Passeriformes	Sturnidae	Acridotheres	Acridotheres tristis
Merops philippinus	Merops philippinus Linnaeus, 1767	Coraciiformes	Meropidae	Merops	Merops philippinus
Copsychus fulicatus	Copsychus fulicatus (Linnaeus, 1766)	Passeriformes	Muscicapidae	Saxicoloides	Copsychus fulicatus
Oriolus kundoo	Oriolus kundoo Sykes, 1832	Passeriformes	Oriolidae	Oriolus	Oriolus kundoo
Dicrurus macrocercus	Dicrurus macrocercus Vieillot, 1817	Passeriformes	Dicruridae	Dicrurus	Dicrurus macrocercus
Dicaeum erythrorhynchos	Dicaeum erythrorhynchos (Latham, 1790)	Passeriformes	Dicaeidae	Dicaeum	Dicaeum erythrorhynchos
Tyto alba	Tyto alba (Scopoli, 1769)	Strigiformes	Tytonidae	Tyto	Tyto alba

Streptopelia decaocto	Streptopelia decaocto (Frivaldszky, 1838)	Columbiformes	Columbidae	Streptopelia	Streptopelia decaocto
Turnix suscitator	Turnix suscitator (J.F.Gmelin, 1789)	Charadriiformes	Turnicidae	Turnix	Turnix suscitator
Ploceus philippinus	Ploceus philippinus (Linnaeus, 1766)	Passeriformes	Ploceidae	Ploceus	Ploceus philippinus
Amaurornis phoenicurus	Amaurornis phoenicurus (Pennant, 1769)	Gruiformes	Rallidae	Amaurornis	Amaurornis phoenicurus
Amandava amandava	Amandava amandava (Linnaeus, 1758)	Passeriformes	Estrildidae	Amandava	Amandava amandava
Lalage melanoptera	Lalage melanoptera (Rüppell, 1839)	Passeriformes	Campephagidae	Coracina	Lalage melanoptera
Pycnonotus cafer	Pycnonotus cafer (Linnaeus, 1766)	Passeriformes	Pycnonotidae	Pycnonotus	Pycnonotus cafer
Gallicrex cinerea	Gallicrex cinerea (Gmelin, 1789)	Gruiformes	Rallidae	Gallicrex	Gallicrex cinerea
Tringa ochropus	Tringa ochropus Linnaeus, 1758	Charadriiformes	Scolopacidae	Tringa	Tringa ochropus
Strix ocellata	Strix ocellata (R.Lesson, 1839)	Strigiformes	Strigidae	Strix	Strix ocellata
Dendrocygna javanica	Dendrocygna javanica (Horsfield, 1821)	Anseriformes	Anatidae	Dendrocygna	Dendrocygna javanica
Psilopogon haemacephalus	Psilopogon haemacephalus (P.L.S.Müller, 1776)	Piciformes	Megalaimidae	Psilopogon	Psilopogon haemacephalus
Amaurornis phoenicurus	Amaurornis phoenicurus (Pennant, 1769)	Gruiformes	Rallidae	Amaurornis	Amaurornis phoenicurus
Amandava amandava	Amandava amandava (Linnaeus, 1758)	Passeriformes	Estrildidae	Amandava	Amandava amandava
Lalage melanoptera	Lalage melanoptera (Rüppell, 1839)	Passeriformes	Campephagidae	Coracina	Lalage melanoptera
Pycnonotus cafer	Pycnonotus cafer (Linnaeus, 1766)	Passeriformes	Pycnonotidae	Pycnonotus	Pycnonotus cafer
Gallicrex cinerea	Gallicrex cinerea (Gmelin, 1789)	Gruiformes	Rallidae	Gallicrex	Gallicrex cinerea
Tringa ochropus	Tringa ochropus Linnaeus, 1758	Charadriiformes	Scolopacidae	Tringa	Tringa ochropus
Strix ocellata	Strix ocellata (R.Lesson, 1839)	Strigiformes	Strigidae	Strix	Strix ocellata
Dendrocygna javanica	Dendrocygna javanica (Horsfield, 1821)	Anseriformes	Anatidae	Dendrocygna	Dendrocygna javanica
Psilopogon haemacephalus	Psilopogon haemacephalus (P.L.S.Müller, 1776)	Piciformes	Megalaimidae	Psilopogon	Psilopogon haemacephalus
Timalia pileata	Timalia pileata Horsfield, 1821	Passeriformes	Timaliidae	Timalia	Timalia pileata
Streptopelia senegalensis	Streptopelia senegalensis (Linnaeus, 1766)	Columbiformes	Columbidae	Spilopelia	Streptopelia senegalensis
Treron phoenicopterus	Treron phoenicopterus (Latham, 1790)	Columbiformes	Columbidae	Treron	Treron phoenicopterus
Sturnia malabarica	Sturnia malabarica (Gmelin, 1789)	Passeriformes	Sturnidae	Sturnia	Sturnia malabarica
Otus sunia	Otus sunia (Hodgson, 1836)	Strigiformes	Strigidae	Otus	Otus sunia
Argya caudata	Argya caudata (Dumont, 1823)	Passeriformes	Leiothrichidae	Turdoides	Argya caudata
Copsychus saularis	Copsychus saularis (Linnaeus, 1758)	Passeriformes	Muscicapidae	Copsychus	Copsychus saularis
Cisticola juncidis	Cisticola juncidis (Rafinesque, 1810)	Passeriformes	Cisticolidae	Cisticola	Cisticola juncidis
Zosterops palpebrosus	Zosterops palpebrosus (Temminck, 1824)	Passeriformes	Zosteropidae	Zosterops	Zosterops palpebrosus
Psittacula krameri	Psittacula krameri (Scopoli, 1769)	Psittaciformes	Psittacidae	Psittacula	Psittacula krameri
Eudynamys scolopaceus	Eudynamys scolopaceus (Linnaeus, 1758)	Cuculiformes	Cuculidae	Eudynamys	Eudynamys scolopaceus
Ploceus manyar	Ploceus manyar (Horsfield, 1821)	Passeriformes	Ploceidae	Ploceus	Ploceus manyar
Dumetia hyperythra	Dumetia hyperythra (Franklin, 1831)	Passeriformes	Timaliidae	Dumetia	Dumetia hyperythra
Picus canus	Picus canus J.F.Gmelin, 1788	Piciformes	Picidae	Picus	Picus canus
Cinnyris asiaticus	Cinnyris asiaticus (Latham, 1790)	Passeriformes	Nectariniidae	Cinnyris	Cinnyris asiaticus
Euodice malabarica	Euodice malabarica (Linnaeus, 1758)	Passeriformes	Estrildidae	Euodice	Euodice malabarica
Artamus fuscus	Artamus fuscus Vieillot, 1817	Passeriformes	Artamidae	Artamus	Artamus fuscus
Ficedula albicilla	Ficedula albicilla (Pallas, 1811)	Passeriformes	Muscicapidae	Ficedula	Ficedula albicilla
Anthus hodgsoni	Anthus hodgsoni Richmond, 1907	Passeriformes	Motacillidae	Anthus	Anthus hodgsoni
Lanius cristatus	Lanius cristatus Linnaeus, 1758	Passeriformes	Laniidae	Lanius	Lanius cristatus

Interpretation: Bird Diversity Across All Sites

1. High Species Richness

- *73 unique species across 11 sites indicates good regional species richness — your broader landscape supports a wide variety of birds representing multiple ecological guilds (insectivores, frugivores, granivores, raptors, and waterbirds).*
- *This suggests the overall region retains heterogeneous habitats — possibly including open woodlands, wetlands, and scrub areas.*

2. Low Species Evenness

- *Each species has a count of 1, meaning individuals were detected rarely or uniformly across sites.*
- *This pattern indicates low abundance and uneven distribution — no dominant species and many rare or occasional ones.*
- *It may reflect sampling limitations (short monitoring duration or low detection probability) or true ecological stress (fragmented habitats, low carrying capacity).*

3. Habitat Fragmentation & Patchiness

- *If each species appears only once across 11 sites, the local (alpha) diversity is likely low even though the regional (gamma) diversity is high.*
- *This points to spatial heterogeneity — species are scattered in isolated pockets instead of being evenly distributed, typical of fragmented or disturbed ecosystems.*

4. Presence of Indicator & Sensitive Species

Species such as:

- *Pitta brachyura (Indian pitta), Hypothymis azurea (Black-naped monarch) → indicate intact forest understorey.*
- *Otus bakkamoena (Indian scops owl), Strix ocellata (Mottled wood owl) → reflect tree cavity availability and low nighttime disturbance.*
- *Dendrocygna javanica (Lesser whistling duck), Tringa ochropus (Green sandpiper) → signal wetland patches.*
- *Merops orientalis (Green bee-eater), Lanius schach (Long-tailed shrike) → indicate open or semi-arid areas.*
- *Their presence confirms that multiple microhabitats coexist within your landscape — a positive sign of ecological mosaic, even if local richness is low.*

5. Human-Associated Species

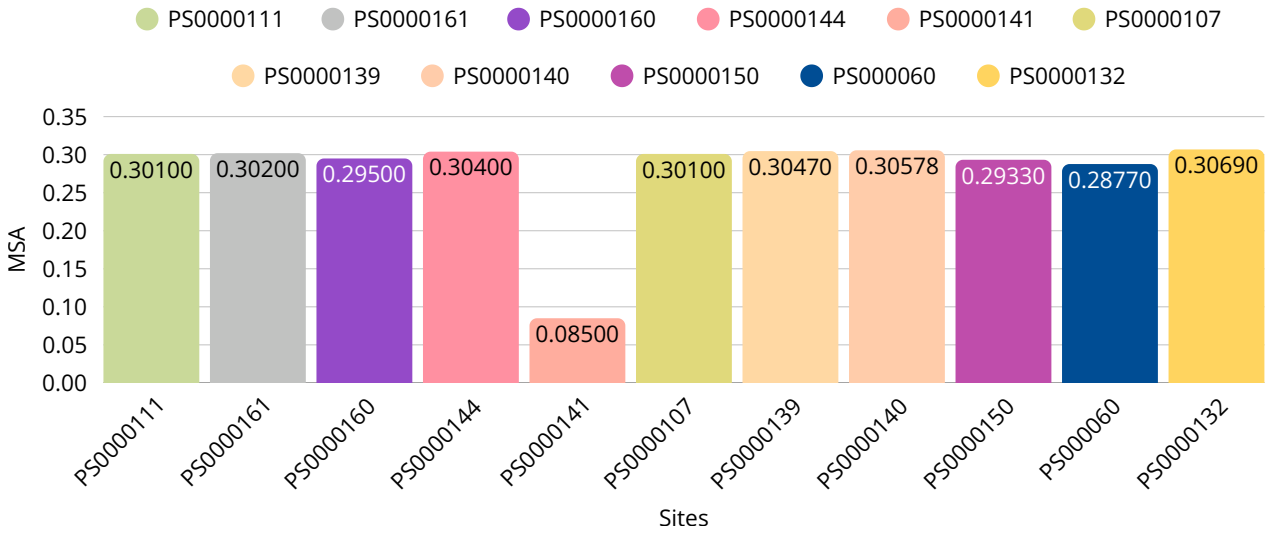
- ***Acridotheres tristis*** *(Common myna),* ***Pycnonotus cafer*** *(Red-vented bulbul),* ***Streptopelia species, Psittacula krameri*** *— these are* ***urban-tolerant and generalist species****.*
- *Their frequent detection (even at low counts) indicates* ***moderate human influence or edge habitats near settlements, agriculture, or roads.***

Mean Species Abundance:

MSA values across the **sites range from 0.085 to 0.307,** indicating generally **low native species** abundance compared to undisturbed reference conditions. Most sites fall around **0.29–0.31, suggesting moderate to high** ecological stress likely due to habitat disturbance, fragmentation, and land-use pressures.

Use of MSA:

- **Project-level biodiversity assessment:** Used to compare restored or managed sites with baseline or control areas.
- **Policy and planning:** Informs land-use planning, environmental impact assessments, and biodiversity offset strategies.
- **Carbon and biodiversity crediting:** Supports nature finance by providing a quantifiable biodiversity metric for integrated biodiversity–carbon projects.
- **SDG and CBD reporting:** Used for national and corporate reporting on biodiversity targets, such as the Kunming–Montreal Global Biodiversity Framework and SDG 15 (Life on Land).



Key Observation

- *The MSA values range from **0.085 to 0.307,** indicating that most sites fall within the "**Very Poor/Collapsed" (≤0.1) and "Poor/Degraded" (0.1–0.3) categories,** with a few reaching the lower bound of **"Heavily Modified/Fair" (0.3–0.5).***
- *Only 3 sites **(PS0000144, PS0000139, PS0000140, PS0000132)** are marginally **above 0.3,** suggesting limited **ecological recovery** or **habitat integrity.***
- *The majority of the landscape is therefore ecologically degraded, with **low native species abundance** and **fragmented habitat conditions.***

Site Code	MSA Value	Interpretation	Score
PS0000111	0.30100	**Moderate habitat condition – balanced species presence with ongoing improvement.**	2
PS0000161	0.30200	**Moderate integrity – species abundance fair, showing gradual recovery.**	2
PS0000160	0.29500	**Early recovery stage – signs of regeneration but still impacted.**	1
PS0000144	0.30400	**Good habitat condition – relatively stable native species abundance.**	3
PS0000141	0.08500	**Severely degraded – major loss of native species; urgent restoration needed.**	0
PS0000107	0.30100	**Moderate condition – noticeable species decline but not severely degraded.**	2
PS0000139	0.30470	**Good condition – stable ecosystem with moderate resilience.**	3
PS0000140	0.30578	**Excellent habitat quality – high native species abundance, resilient system.**	4
PS0000150	0.29330	**Early recovery – improving diversity but still under pressure.**	1
PS000060	0.28770	**Severely degraded – very low native abundance, intensive recovery required.**	0
PS0000132	0.30690	**Excellent condition – high MSA, well-functioning ecosystem.**	4

Recommendations:

- ***Critically Low MSA Sites (MSA < 0.29; e.g., PS0000141, PS000060):*** *Prioritize deep soil recovery with organic amendments and microbial inoculants before planting.*
- ***Moderate MSA Sites (MSA > 0.29; e.g., PS0000111, PS0000161):*** *Continue with native species planting while implementing strict controls on grazing and agricultural expansion to reduce pressure on native species.*

Species

Conservation Value

The species recorded across the project sites have been categorized based on the International Union for Conservation of Nature (IUCN) Red List classifications, which indicate their relative risk of extinction

IUCN Category	Number of Species	Description
Critically Endangered (CR)	**0**	Facing an extremely high risk of extinction in the wild.
Endangered (EN)	**0**	At a very high risk of extinction due to ongoing threats.
Vulnerable (VU)	**0**	High risk of becoming endangered unless the circumstances improve.
Near Threatened (NT)	**1**	Close to qualifying for a threatened category and require monitoring.
Least Concern (LC)	**72**	Currently widespread and abundant, though continued habitat loss is a risk.

Interpretation:
The site has high conservation status because of the presence of **Alexandrine Parakeet,** a **Near threatened** species**.**

Recommendations:

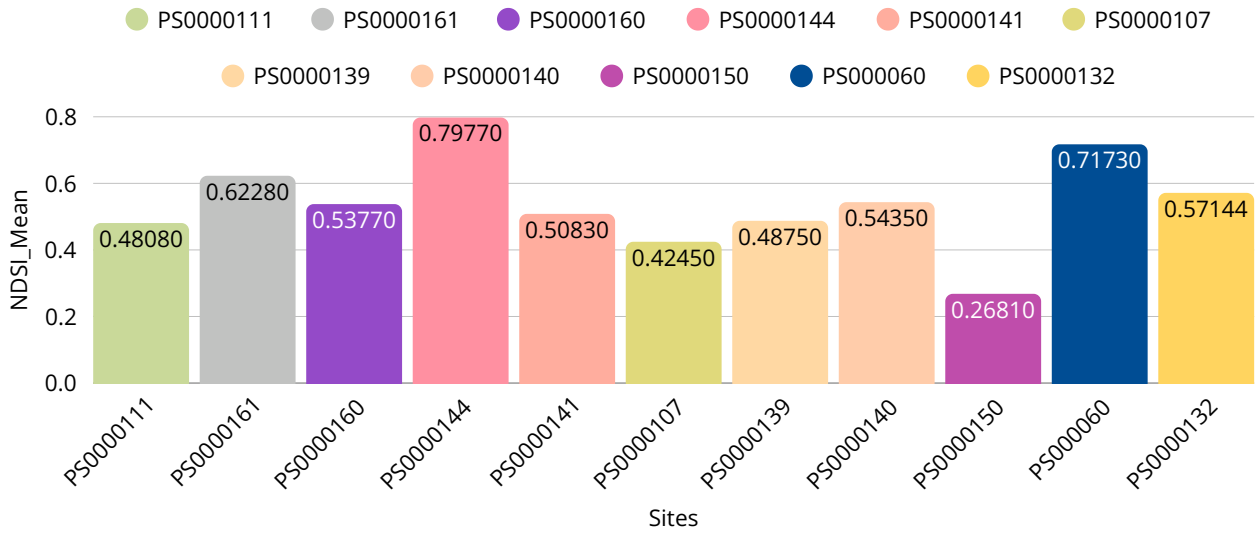
- ***Monitor Near-Threatened Species****: Regularly track the* ***NT species to detect early signs of decline.***
- ***Protect Habitat:*** *Maintain and restore key habitats to prevent fragmentation and degradation.*
- ***Long-Term Monitoring:*** *Use bioacoustics, camera traps, and surveys to track overall biodiversity trends.*

The Normalized Difference Soundscape Index (NDSI):

The Normalized Difference Soundscape Index (NDSI) quantifies the balance between biophony (natural biological sounds such as birds, insects, and amphibians) and anthrophony **(human-made sounds such as traffic, industry, and machinery)** within an environment.

Uses of NDSI:

- **Assess Habitat Quality:** Higher biophony relative to anthrophony indicates healthier, less disturbed ecosystems.
- **Monitor Human Disturbance:** Detects the impact of urbanization, industrial activity, and noise pollution on wildlife.
- **Biodiversity Monitoring:** Serves as a proxy for species activity and richness in bioacoustic surveys.
- **Conservation Planning:** Helps identify priority areas for protection and management based on soundscape integrity.



Key Insights:

1. *Mean NDSI Values :*

- *Range: 0.268 – 0.798, indicating low/moderate human/noise pressure across sites (higher values = less anthropogenic noise).*
- *Most sites cluster around 0.42 – 0.62, suggesting moderate suppression of biological sounds.*
- *The highest site value (0.798, PS0000144) experiences least human/noise influence.*
- *The lowest site value (0.268, PS0000150) represents relatively high human disturbance.*

2. ***Health Classification (health_class):***

- *All sites are classified as "Moderate (0.2–0.8)", meaning human noise is present but not extreme, and biological sounds are detectable.*

Site	NDSI	Interpretation	Recommendation
PS0000111	0.48080	**Early Recovery**	Frequent monitoring. Implement active, localized noise mitigation (e.g., vegetation buffers, seasonal restrictions) to encourage biophony.
PS0000161	0.62280	**Excellent**	Protect site; minimal intervention needed. Monitor intermittently to ensure noise remains low.
PS0000160	0.53770	**Good**	Protect site; minimal intervention needed. Monitor intermittently to ensure noise remains low.
PS0000144	0.79770	**Excellent**	Reference site quality. Rigorously protect from any new noise sources. Minimal monitoring required.
PS0000141	0.50830	**Moderate**	Regular monitoring. Implement minor, targeted mitigation measures to reduce intermittent human disturbance where feasible.
PS0000107	0.42450	**Early Recovery**	Frequent monitoring. Implement active, localized noise mitigation (e.g., vegetation buffers, seasonal restrictions) to encourage biophony.
PS0000139	0.48750	**Moderate**	Regular monitoring. Implement minor, targeted mitigation measures to reduce intermittent human disturbance where feasible.
PS0000140	0.54350	**Good**	Protect site; minimal intervention needed. Monitor intermittently to ensure noise remains low.
PS0000150	0.26810	**Severely Degraded**	Prioritize immediate, aggressive noise management interventions (e.g., traffic control, acoustic barriers) to restore soundscape quality.
PS0000060	0.71730	**Excellent**	Reference site quality. Rigorously protect from any new noise sources. Minimal monitoring required.
PS0000132	0.57144	**Excellent**	Protect site; minimal intervention needed. Monitor intermittently to ensure noise remains low.

Recommendations:

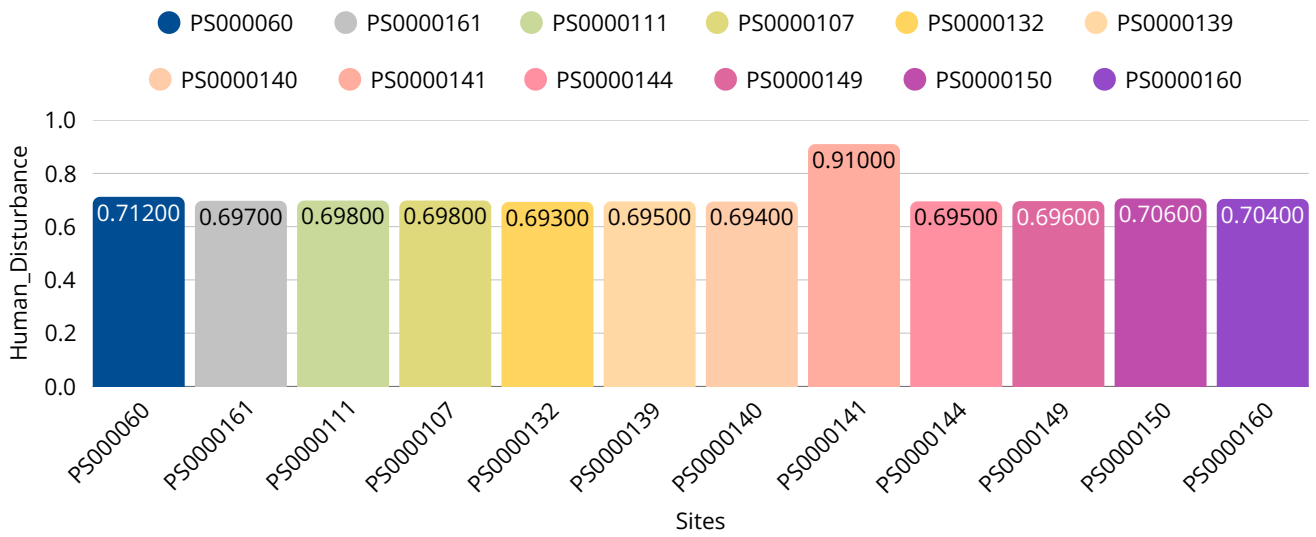
- ***High Noise & Disturbance Sites:*** *Prioritize immediate noise management, including planting vegetative buffers and restricting machinery use.*
- ***Low Noise/Disturbance Sites:*** *Maintain current conditions and protect these sites as valuable reference areas.*

Human Disturbance:

Human disturbance refers to the degree of pressure or impact caused by human activities on ecosystems, habitats, and species. It includes direct activities such as deforestation, agriculture, grazing, fuelwood collection, infrastructure development, and settlements, as well as indirect effects like noise, pollution, and habitat fragmentation.

Importance of Measuring Human Disturbance

- **Indicator of habitat health** – High disturbance usually means loss of species richness, reduced vegetation cover, and ecosystem degradation.
- **Guides restoration efforts –** Helps identify priority sites for urgent intervention versus areas that need protection.
- **Monitors recovery** – Tracking disturbance over time shows whether restoration efforts are reducing human pressure.
- **Supports policy & management** – Provides evidence for regulating human activities like grazing, logging, or land-use change.



Key Intervention:

- ***Habitat Protection:*** *Implement buffer zones and restrict high-impact activities like grazing, logging, and construction near sensitive areas.*
- ***Community Engagement:*** *Work with local communities to promote sustainable resource use and raise awareness about the ecological value of the sites.*
- ***Restoration and Regeneration:*** *Strengthen vegetation cover through native tree plantations (e.g., mango, arjun, sonajhuri, sagwan) to reduce soil erosion and create wildlife refuges.*
- ***Monitoring & Enforcement:*** *Use periodic field surveys and bioacoustic monitoring to track human activity and enforce site-specific protection measures.*
- ***Alternative Livelihoods:*** *Encourage eco-friendly income options for local populations to reduce dependence on forest resources.*
- ***Noise & Pollution Control:*** *Limit industrial, traffic, and other anthropogenic noise near critical habitats to maintain natural soundscape balance.*

Site Code	Value	Interpretation	Score
PS000060	0.71200	**High Human Disturbance**	1
PS0000161	0.69700	**High Human Disturbance**	1
PS0000111	0.69800	**High Human Disturbance**	1
PS0000107	0.69800	**High Human Disturbance**	1
PS0000132	0.69300	**High Human Disturbance**	1
PS0000139	0.69500	**High Human Disturbance**	1
PS0000140	0.69400	**High Human Disturbance**	1
PS0000141	0.91000	**Low Human Disturbance**	4
PS0000144	0.69500	**High Human Disturbance**	1
PS0000149	0.69600	**High Human Disturbance**	1
PS0000150	0.70600	**High Human Disturbance**	1
PS0000160	0.70400	**High Human Disturbance**	1

Recommendation:

Most sites exhibit high human disturbance, mainly from ***nearby agriculture, settlements, and land-use activities. Establishing vegetative buffers, promoting community-based sustainable practices, restricting access in sensitive*** *areas, and improving habitat connectivity will* ***help reduce human pressure, restore ecological balance, and strengthen long-term ecosystem resilience.***

Conclusion

The Dumka Agroforestry Project begins from a position of high potential but moderate constraint. The Year-0 baseline confirms a **strong biotic foundation: 73 unique bird species** and **moderate acoustic health (0.473)**, indicating a robust regional species pool and high ecological recovery potential.

However, the sites are in an "**early recovery band**" (Biodiversity Score of 22/44). Low vegetation health (NDVI 0.388) and moderate human disturbance confirm that habitat structure (canopy, connectivity) and population abundance (MSA 0.085–0.307) are the immediate binding constraints to achieving full nature-positive outcomes.

Restoration Priorities and Measurable Targets

To address these constraints and accelerate uplift, the strategy is clear and measurable:

Accelerate Habitat Quality: Implement diverse native plantings to improve canopy development and vertical structure.

Restore Connectivity: Stitch sites together using riparian buffers and native hedgerows to enhance movement pathways and raise habitat connectivity.

Mitigate Threats: Implement access management and quiet-hour protocols to reduce mid-day noise and anthropological pressures.

Targeted Metrics: Achieve measurable gains in 12–24 months, including raising NDVI by +0.04 and increasing acoustic health to ≥ 0.6 at priority sites.

By focusing on these targets, the project will lift the Ecosystem Condition and Population Status scores, securing the existing strength of the Species Assemblage pillar. **This evidence-backed approach provides an auditable foundation aligned with TNFD/LEAP expectations, coupling carbon revenues with transparent, statistically demonstrable biodiversity recovery and community co-benefits across the Dumka restoration mosaic.**

Darukaa.earth

CONTACTS

Website: darukaa.earth
Email: vikram.singh@darukaa.com
Mobile: +91 99873 68383